

Targeting Against Shocks: Data, Experts, or Impact-based Targeting in Lebanon*

Ghassan Baliki[†] Micah Cruz[‡] Felipe Dunsch[‡] Jonathan Garcia[§]
Andrea Guariso[¶] Jonas Heirman[‡] Dorothee Weiffen[†]

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Abstract

Humanitarian funding is increasingly volatile, forcing actors to decide which households continue receiving assistance when budgets are cut, yet little is known about how alternative targeting approaches perform at this tight margin. This paper experimentally compares an expert-driven approach designed by policymakers with three data-driven methods for prioritizing beneficiaries during the scale-down of a nationwide food assistance program in Lebanon, shortly before an unexpected conflict escalation. The four methods produce systematically different beneficiary profiles but remarkably similar average welfare, as they largely agree on the poorest households and differ mainly over marginal cases. The expert approach elicits more complaints, ranks deprivation least accurately, and delivers lower welfare than the data-driven methods. Exploiting the randomized assignment and the subsequent conflict escalation, we find that continued assistance mitigated the deterioration in psychosocial well-being but not the material consequences of conflict. Households prioritized before the conflict under the Food Consumption Score (FCS) targeting method were less likely to be displaced and to deplete productive assets. Finally, we compare deprivation-based targeting with an impact-based approach that prioritizes households by their predicted welfare loss from assistance withdrawal. The two overlap on only 65% of selected households, yet the impact-based rule captures about 90% of the maximum achievable welfare, compared with 60–70% for the deprivation-based methods, at no additional data-collection cost.

Keywords: targeting, humanitarian aid, shocks, proxy means tests, food security, phase-out, conflict, machine learning, social welfare.

JEL codes: H53, I38, O12, Q18.

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[†]International Security and Development Center (ISDC).

[‡]World Food Programme.

[§]Humboldt University of Berlin and IGZ.

[¶]University of Milan-Bicocca.

1 Introduction

Humanitarian aid is intended as a temporary instrument that supports households in meeting their basic needs, often in response to a shock, until conditions improve and assistance is no longer required (Jeong and Trako, 2022). Its funding also comes from streams that are often volatile, shaped by changes in resource availability and shifts in donor priorities (Sterck and Bruni, 2025; Kagin et al., 2024; Chawla, 2025). These features make humanitarian targeting distinctive in two ways: it must identify both chronic deprivation and shock-induced vulnerability, and it must inform decisions not only about program entry but also about how assistance is reduced as needs decline or resources tighten. Yet most targeting tools and evidence are drawn from social protection programs, which are designed to identify households facing persistent income constraints (Hanna and Olken, 2018). Evidence also focuses largely on targeting at program entry. As a result, little is known about whether existing approaches can identify both chronically and transitorily vulnerable households, or which methods are best suited to determine who should remain eligible at the exit margin.

Data-driven methods such as proxy means tests (PMTs) are among the most common approaches for allocating resources in humanitarian contexts (Grosh and Baker, 1995; Altındağ et al., 2021). PMTs use observable household characteristics, such as assets and demographic variables, to predict welfare. Their appeal traces back to social protection, where they are seen as objective, standardized, and resistant to manipulation (Follett and Henderson, 2023). However, PMTs were largely developed and validated against relatively persistent measures of welfare, such as poverty, rather than more volatile indicators of deprivation, such as food insecurity (Mills et al., 2015; Schnitzer and Stoeffler, 2024). This raises concerns about their ability to identify acute need and rapidly changing vulnerability in humanitarian crises.

In this paper, we leverage the evaluation by Baliki et al. (2025) of the phase-out of a nationwide social protection program, during which households faced two unexpected shocks: the cut in humanitarian assistance and the Israeli ground invasion of Lebanon. We compare four data-driven targeting approaches, assessing both their targeting performance and their ability to mitigate the welfare consequences of these two shocks, assistance withdrawal and conflict exposure. First, we draw on the initial randomization across the four approaches to compare their performance, which households each one selects and the welfare effects of losing assistance under each. Second, we exploit the escalation of conflict between survey rounds to examine whether the impact of conflict intensity on displacement, household assets, and other welfare indicators was moderated by continued receipt of assistance and by the targeting method through which households were selected. Finally, we use the assistance-withdrawal shock to develop a fifth, impact-based targeting approach that ranks households by their predicted welfare losses from losing assistance rather than their observed baseline vulnerability. We compare its performance and cost-effectiveness with that of the four randomized targeting methods.

We examine the first question by comparing the four targeting approaches designed by Baliki et al. (2025). The evaluation included three proxy means test methods that identify vulnerable households using different welfare criteria: expenditure (PMT-Expenditure), the Food Consumption Score (PMT-FCS), and the likelihood of food insecurity based on the composite Consolidated Approach for Reporting Indicators of Food Security (CARI) indicator (PMT-

CARI).¹ Each method reflects a distinct conception of need. The expenditure-based PMT targets chronic poverty, the FCS-based PMT captures food deprivation, and the CARI-based PMT identifies broader vulnerability by combining food security outcomes with coping capacity. All PMTs are calibrated using the nationally representative 2023 Lebanese Household Survey (LHS). The fourth method is an Expert method based on eligibility criteria derived from a survey of national experts, who identified the household characteristics most indicative of need for food assistance.

We begin by evaluating the sensitivity of targeting scores to alternative modeling choices, given the widespread use of expenditure-based PMTs, which benefit from a continuous welfare measure and well-established benchmarks such as poverty lines. Across expenditure, FCS, and the CARI, the least absolute shrinkage and selection operator (LASSO) performs at least as well as OLS and generalized random forests (GRF), suggesting limited gains from more flexible machine-learning methods. We also examine which information drives these predictions.² For all three PMTs, roughly half of predictive power comes from socioeconomic conditions (46–53%), about a third from durable assets (26–32%), and the remainder from demographics. The relative importance of these components nevertheless varies across PMTs: the expenditure-based PMT places greater weight on demographics, the FCS-based PMT on socioeconomic conditions, and the CARI-based PMT falls in between.

We then evaluate predictive accuracy using the training data (LHS), assessing both classification at the 43% eligibility threshold and ranking performance across the full distribution. The expenditure-based PMT performs best across all measures, achieving the highest precision (69%)³ and the most accurate ranking of households (AUC = 0.83).⁴ PMT-CARI (precision = 62%, AUC = 0.77) and PMT-FCS (precision = 58%, AUC = 0.71) also perform well, while the Expert method performs substantially worse (precision = 55%, AUC = 0.67). All methods nonetheless outperform random targeting (precision = 41%, AUC = 0.50) by a considerable margin. We also find that all methods are progressive, though to different degrees. Relative to an asset index in the target population, a one-standard-deviation increase in assets reduces the odds of selection by 87% under PMT-FCS, 83% under PMT-CARI, 77% under the expenditure-based PMT, and 66% under the Expert method, indicating that the latter is least effective at prioritizing asset-poor households.

Next, we study whether the methods lead to the selection of different beneficiary profiles.

¹CARI is a harmonized WFP methodology that combines individual-level food security indicators to classify households into four groups—food secure, marginally food secure, moderately food insecure, and severely food insecure—consistent with the Integrated Food Security Phase Classification (IPC). See the WFP CARI guidance manual: <https://vamresources.manuals.wfp.org/docs/the-consolidated-approach-for-reporting-indicators-of-food-security-cari>.

²The covariates each LASSO retains are classified into durable assets, demographic characteristics, and socioeconomic conditions, and each is weighted by its contribution to the model’s prediction. A covariate’s contribution is the absolute value of its penalized LASSO coefficient multiplied by the covariate’s standard deviation, a standard normalization that places predictors measured on different scales on a comparable footing. These contributions are summed within each group and expressed as a share of the model total, so the shares sum to 100% within each PMT and remain comparable across models despite differences in the selected penalty.

³Precision is the share of selected households that are truly eligible (below or above the 43% threshold, depending on the outcome) (Aiken et al., 2022).

⁴The area under the ROC curve (AUC) summarizes ranking across all thresholds and equals the probability that a randomly chosen eligible household is ranked above a randomly chosen ineligible one (0.5 = random, 1 = perfect) (Verme and Gigliarano, 2019).

We find that they do, and that the starkest contrast is between the Expert method and the PMTs: the Expert method selects much older household heads (62 versus 52 years under PMT-Expenditure), far more female-headed (54% versus 25%) and monoparental households, more households with no employed member (60% versus 28%), and smaller households with fewer children. Differences among the three PMT methods are also evident. PMT-FCS stands apart while PMT-Expenditure and PMT-CARI are more alike. Relative to PMT-Expenditure, PMT-FCS selects smaller households (4.0 versus 4.7 members), with older heads (55 versus 52 years), more female-headed (32% versus 25%), and more often with no employed member (36% versus 28%). On the asset index, both PMT-FCS and PMT-CARI reach somewhat poorer households than PMT-Expenditure (around -0.65 versus -0.50 standard deviations).

These differences in selected profiles, model inputs, and accuracy are reflected in substantial variation in who is selected by each method. The strongest overlap occurs between PMT-CARI and PMT-Expenditure, yet their pairwise correlation is only 0.65. The remaining PMT pairs agree even less (PMT-FCS with PMT-CARI at 0.59 and with PMT-Expenditure at just 0.41), and the Expert method is almost orthogonal to the PMTs (0.02 to 0.18). Across all six pairs the average correlation is only 0.33, rising to 0.55 among the three PMTs alone, so any two methods select largely distinct sets of households.

Having established differences in accuracy and beneficiary profiles across methods, we examine how these translate into satisfaction—proxied by appeals—and other downstream outcomes following the discontinuation of assistance at the first round of data collection. We study outcomes not explored by [Baliki et al. \(2025\)](#) and seek explanations for any differences across methods. The Expert method produced higher appeal rates, 6 p.p. above the average appeal rate of 50% among discontinued households. This result is in line with the Expert method selecting less asset-poor households on average, and is consistent with concerns about coarse targeting criteria and perceived legitimacy.

When looking at welfare outcomes, we find that assignment to a particular targeting approach generates no differences in expenditure, FCS, or well-being indicators, even when we restrict the sample to beneficiary households. This null is informative. The methods select different households, but the households they swap at the margin have similar average outcomes. Method choice therefore matters more for who is included than for the average welfare of those included. The four methods often select different households, with only about one-third of beneficiaries overlapping across methods. Yet these differences have virtually no effect on average downstream welfare, because the methods agree on the poorest households and differ only over a marginal group whose gains from assistance are close to the caseload average. As a result, method choice matters more for who receives assistance than for the average welfare it delivers.

We leverage the unexpected escalation of the conflict during the evaluation, combined with the experiment’s randomization, to examine the effects of this shock on households and to assess whether receiving assistance and the randomly assigned targeting method mitigated its impacts. Exposure was widespread, affecting 56% of households during the study period. Although concentrated in the south, exposure was unrelated to targeting-arm assignment or to whether a household was still receiving assistance before the conflict intensified. Households across the four arms were equally likely to be exposed and experienced similar conflict intensity ($p = 0.23$).

Therefore, we compare changes in household outcomes before and after the conflict escalation in relation to conflict intensity, examining how these changes differ by assistance receipt and the assigned targeting approach.

The shock was severe. A one-standard-deviation increase in conflict intensity was associated with a 23 percentage-point (p.p.) higher probability of displacement and 18 p.p. more shelter damage, sharply higher trauma, and a durable-asset decline of about 0.14 standard deviations. Continued assistance was associated with a roughly one-quarter reduction in the effect of conflict on trauma. The targeting method also shaped how households absorbed the shock, with those assigned to PMT-FCS distinctly better protected than those targeted on expenditure. Among households that kept their assistance, PMT-FCS selection cut the probability of displacement by 7.5 p.p. and of shelter damage by 5 p.p. as conflict intensity rose, and across the full population it lowered displacement by 2.5 p.p. and sales of productive assets by 4.6 p.p. Households selected on food insecurity were therefore better insulated from the conflict than those selected on expenditure.

The four targeting methods prioritize households deemed most vulnerable at the time of selection. The subsequent escalation of conflict illustrates that vulnerability is dynamic. Even assisted households were adversely affected, and the targeting method itself, PMT-FCS in particular, shaped how households experienced the shock. We therefore examine whether targeting based on expected impacts would lead to different outcomes than targeting based on observed vulnerability. To do so, we build two impact-based targeting rules using the identification strategy in [Baliki et al. \(2025\)](#). We estimate heterogeneous effects of assistance discontinuation on total and food expenditure, the two outcomes for which meaningful treatment-effect heterogeneity is found. Following the CATE-ranking framework of [Chernozhukov et al. \(2018\)](#), as applied by [Haushofer et al. \(2025\)](#), households are ranked according to their predicted conditional average treatment effects (CATEs), from those predicted to suffer the largest losses from assistance withdrawal to those expected to suffer the least. Unlike [Haushofer et al. \(2025\)](#), however, we treat the withdrawal rather than the introduction of assistance as the relevant shock.

We find substantial divergence between impact- and vulnerability-based selection. Only 65% of households overlap when targeting is based on total expenditure, and just 61% when targeting is based on food expenditure. While the proxy-means tests concentrate on asset-poor households, the impact-based approach selects households with roughly the same level of assets as the average beneficiary but a much higher share of elderly-headed households. This suggests that, despite not appearing more deprived in terms of assets at the time of the survey and therefore being less likely to be selected by the proxy-means tests, elderly-headed households were more vulnerable to the loss of assistance and were therefore less likely to be identified by deprivation-based targeting.

Welfare simulations suggest that targeting on impact captures a far larger share of the attainable welfare gain—about nine-tenths of the first-best on expenditure, against six- to seven-tenths for the deprivation-based methods and least of all for the expert rule—yet it selects households that are no poorer than average on assets. The households most deprived today are not those who lose the most when assistance is withdrawn. This advantage is specific

to the expenditure outcomes the rule targets: it does not extend to food security or to the conflict outcomes, where ranking on expenditure loss buys no welfare over the deprivation-based methods.

Combining the welfare simulations with back-of-the-envelope implementation costs does not alter the conclusions. Screening costs account for only 1.5 to 4.4% of the transfer budget. The proxy-means tests remain more cost-effective than the Expert rule, even if the latter could rely on an existing population registry rather than a full census. Although such a registry would reduce implementation costs, the savings would not offset the welfare losses from less effective targeting. Under the same assumptions, the simulated impact-based rule would be even more cost-effective than the vulnerability-based PMTs.

By comparing the trade-offs across alternative data-driven targeting approaches, including impact-based methods that leverage negative shocks, we contribute to the literature on beneficiary targeting on multiple fronts.

First, we add to the scarce literature on how to effectively reach the most vulnerable in fragile settings (Jeong and Trako, 2022), and we extend recent work on data-driven targeting, including Altındağ et al. (2021) on Syrian refugees in Lebanon and Aiken et al. (2022). We show that simple regularized regression predicts welfare as well as more flexible machine learning, and that although the four methods select largely distinct beneficiaries, they deliver similar average welfare because they agree on the poorest and differ only over marginal households.

Second, we contribute to the scarce evidence on the overlap between poverty-targeted programs and those designed to enhance food security, as most studies focus on anti-poverty programs (Coady et al., 2004; Grosh et al., 2022; Hastings and Washington, 2010), offering limited guidance for targeting in shock-affected or humanitarian contexts, where vulnerabilities are often more transient (Premand and Schnitzer, 2021) and outcomes are affected by seasonality (Mills et al., 2015). Comparing expenditure- and food-security-based tests, we find they reach overlapping but distinct households and deliver similar average welfare, yet food-security targeting (PMT-FCS) leaves households somewhat better insulated from the conflict shock.

Third, we are among the few studies to compare selection approaches driven by expert judgment, as most existing evidence focuses on leaders or community-based agents whose decisions may be shaped by corruption, elite capture, nepotism, and other distortions (Aiken et al., 2025; Alatas et al., 2012; Basurto et al., 2020; Sumarto et al., 2025; Trachtman et al., 2025). We find the Expert rule selects a categorically vulnerable population but ranks the deprived least accurately, attracts the most appeals, and delivers the lowest welfare of the methods.

Fourth, by examining outcomes beyond standard precision metrics, we provide evidence on broader trade-offs that are highly relevant in humanitarian settings, including satisfaction and welfare outcomes (Hillebrecht et al., 2020; Premand and Schnitzer, 2021; Sabates-Wheeler et al., 2015; Schnitzer and Stoeffler, 2024). Beyond accuracy, we are among the few studies measuring the downstream outcomes that could be affected by the targeting approach, such as Alatas et al. (2012) and Premand and Schnitzer (2021).

We contribute to the emerging literature on the protective role of humanitarian assistance against adverse shocks by examining how its effectiveness varies across targeting methods. Existing evidence shows that cash transfers help cushion households against covariate shocks

such as drought and conflict (Premand and Stoeffler, 2022; Knippenberg and Hoddinott, 2017; Asfaw and Davis, 2018; Otchere and Handa, 2022; Caeiro et al., 2025), yet much less is known about whether this protective effect depends on who is targeted. The closest evidence comes from studies comparing deprivation- and impact-based targeting outside humanitarian crises (Haushofer et al., 2025) and from recent work showing that incorporating information on shocks into targeting can reduce exclusion errors during economic downturns (Beuermann et al., 2025). We provide some of the first experimental evidence on whether targeting based on food security, which captures more transient deprivation than chronic poverty, improves households’ resilience to adverse shocks.

We also extend the emerging literature on impact-based targeting to humanitarian programs. Previous studies have contrasted targeting based on predicted impacts rather than deprivation in settings such as selecting the poorest of the poor versus those most likely to succeed (Haushofer et al., 2025), minimum wage reforms and self-targeting in public works programs (Crépon et al., 2026), and job-search assistance for refugees (Caria et al., 2024). We instead target households according to their predicted welfare losses from assistance discontinuation. Evidence of this kind is rare because deliberately withdrawing assistance raises ethical concerns, leaving few settings in which the consequences of program phase-out can be studied directly. By exploiting the innovative design of Baliki et al. (2025), which generates exogenous variation in assistance discontinuation, we provide unique experimental evidence on this targeting trade-off for humanitarian phase-out.

The remainder of the paper is structured as follows. Section 2 describes the context, the in-kind assistance program implemented by WFP, and the four targeting approaches. Section 3 outlines the study design and data. Section 4 discusses internal validity, the empirical strategy, and the welfare framework used to compare deprivation- and impact-based targeting. Section 5 reports the main findings, covering targeting performance, the effects of assistance discontinuation, and the protective role of assistance during the conflict. Section 6 concludes.

2 Setting and Intervention

2.1 Setting

In recent years, Lebanon has faced one of the most severe crises in its modern history, with GDP per capita falling by more than 50% between 2019 and 2021 and the country being downgraded to lower-middle-income status after 25 years as an upper-middle-income country (World Bank, 2024). The economic crisis, triggered by the 2019 financial collapse and compounded by the August 2020 Beirut Port explosion and the COVID-19 pandemic, sharply eroded households’ purchasing power and increased food insecurity (IPC, 2025; World Bank, 2024). These pressures have unfolded in a context where Lebanon hosts an estimated 1.5 million displaced Syrians, nearly 64% of whom live below the survival poverty line (Verme et al., 2015), further straining scarce resources and deepening social tensions and inequalities between host and refugee communities.

These compounded crises prompted the government and humanitarian agencies, including the World Food Programme (WFP), to expand support to the Lebanese population. This included increasing the number of beneficiaries assisted through existing government programs, as well as

introducing new interventions, such as the in-kind support program launched in 2020 by WFP in coordination with the Ministry of Social Affairs (MoSA). The program provided emergency food assistance to vulnerable households through monthly food parcels and gradually expanded to cover up to 60,000 households by 2023, equivalent to around 5% of the Lebanese population.⁵

By 2023, a growing number of competing humanitarian emergencies were drawing donor funding away from protracted crises such as Lebanon toward newer emergencies, leaving needs in the country increasingly ahead of available resources (WFP, 2025). Against this backdrop of funding shortfalls and shifting donor priorities, WFP had to scale back its operations and reassess which households should remain eligible for assistance, reducing its caseload by more than 40% to a final number of around 25,000 households.

In late September 2024, long-standing tensions with Israel escalated sharply along the southern border, triggering the deadliest period in Lebanon since 2006 (WFP, 2025). The conflict spread well beyond South Lebanon, causing widespread fatalities and injuries and displacing some 1.6 million people—including roughly 900,000 internally displaced—before a ceasefire on 27 November 2024 (WFP, 2025). The escalation pushed acute food insecurity to 29% of the population (WFP, 2025).

2.2 In-kind Humanitarian Program

WFP launched its in-kind assistance program for Lebanese residents in 2020 in response to the country’s overlapping crises, which had led to a sharp deterioration in food security. Beneficiaries of the program received support for the duration of a 12-month “cycle,” with renewal conditional on funding availability.⁶ Each household was entitled to one food parcel for every two members, up to a maximum of three parcels. Each parcel contained around 20 kg of food staples, valued at USD 32 in January 2023, equivalent to about 30% of median monthly food expenditure and 13% of total household expenditure.⁷ The parcels were designed to provide roughly 1,300 kcal per person per day, as detailed in Appendix Table A1.

The program expanded progressively from 2020 onwards and was implemented in close collaboration with Lebanon’s national social protection system. By 2023, WFP was providing in-kind food assistance to 58,501 Lebanese households across the country,⁸ as shown in Appendix Figure A1. Such collaboration with national systems is a common feature of WFP’s operations globally, where the organization provides financial and technical support to facilitate both the vertical and horizontal expansion of assistance, particularly in times of crisis. A central challenge in these partnerships is the alignment of targeting strategies. WFP’s mandate is to support food-insecure households—a condition that is often more transient and shock-driven (Jones et al., 2013)—while national social protection programs typically focus on poverty or other chronic vulnerabilities. An important question is what the implications are of emphasizing WFP’s objectives versus those of the government when determining targeting approaches.

⁵ Author’s estimation based on the number of households in the program and average household size.

⁶ Between 2020 and 2023 coverage of the program was progressively expanded and assistance was continuously renewed.

⁷ Authors’ estimation using the nationally representative Lebanon Household Survey (LHS) data.

⁸ An additional 15,019 households were reached by WFP with food assistance. These households were however enrolled in the program at different stages through different procedures and were not part of the re-targeting exercise and, hence, of this study.

For the in-kind program, WFP used a mix of approaches and outcomes to identify beneficiaries. Initially, it relied on lists of vulnerable households provided by community partners and the government, complemented by self-registration through an online portal or hotline. Eligibility was then validated through household surveys. In the first rounds, selection was based on Food Consumption Scores (FCS)⁹ collected by phone, while in later rounds inclusion criteria shifted to WFP’s Multidimensional Deprivation Index (MDDI)¹⁰ measured through in-person surveys.

By 2023, funding shortfalls and shifting priorities led to a reduction in the beneficiary caseload for the 2024 cycle, from 58,501 to 25,160 households (43%¹¹ of the 2023 level). The funding cuts created an opportunity to evaluate alternative targeting approaches through a randomized experiment. Following a beneficiary census conducted by WFP between August and October 2023, beneficiaries were randomly assigned to one of four targeting methods, each of which retained the 43% most vulnerable households as eligible for assistance. Households were notified of their eligibility by SMS in December 2023, before the 2024 assistance cycle began in January 2024.

Although the 2024 cycle initially reflected the funding-driven reduction in beneficiaries, by August 2024 WFP had begun scaling up assistance as the conflict with Israel intensified. Following Israel’s ground invasion, it expanded its emergency operations using a Conflict Impact Index to prioritize the worst-affected cadasters. Registered households in these priority areas that were not already receiving assistance received temporary emergency cash transfers, reaching around 62,000 conflict-affected Lebanese households (WFP, 2025).

2.3 Targeting Approaches

In our study, all households deemed eligible receive the same food basket; what varies is the method used to determine eligibility. Eligibility is determined using one of four targeting approaches evaluated by Baliki et al. (2025): three proxy means test (PMT) models and one Expert method. These approaches reflect WFP’s prior targeting practice, address open questions in the literature, and speak to policy concerns about the relative performance of methods designed to identify chronic versus transient vulnerability.

The PMT models were calibrated and trained using the Lebanon Household Survey (LHS), a nationally representative dataset collected between January and May 2023 that contains the three target outcomes: per-capita expenditure, FCS, and CARI. The models were tested against the 126 predictors common to the LHS and the WFP beneficiary census, which covered 58,501 households in August and September 2023. Covariates were selected by machine learning to

⁹The FCS is a composite indicator used by the World Food Programme. It combines data on a household’s dietary diversity, consumption frequency over the past 7 days, and the relative nutritional importance of different food groups to produce a single score that serves as a proxy for household food security and caloric adequacy (WFP, 2008).

¹⁰The Multidimensional Deprivation Index (MDDI) is a household-level measure of non-monetary poverty based on deprivations in six essential needs dimensions: food, health, education, shelter, WASH, and safety (WFP, 2023).

¹¹The 43% target was based on WFP’s estimation of the number of beneficiaries with consumption levels below the minimum expenditure basket (MEB), which was set at LBP 3,238,315 or USD 71 per capita per month (January 2023 prices). Such estimation was originally based on extrapolation from the nationally representative LHS dataset. Notably, this figure aligned closely with findings from the post-distribution monitoring survey that WFP conducted in the first half of 2023 among a representative sample of beneficiaries, which indicated that 44% of beneficiary households were below the MEB threshold.

maximize predictive power, coefficients were estimated on the LHS, and the models were applied to the census to predict each outcome from the common covariates.

PMT targeting per-capita expenditure (PMT-Exp, T1). Beneficiaries are identified through an econometric PMT model designed to identify households with lower per-capita monthly expenditure. The PMT model predicts household monthly per-capita expenditure from more than 100 expenditure items, both food and non-food. The food items capture purchases, transfers, and own production. Expenditure-based PMT is the most common approach in low- and middle-income settings, where direct poverty measures are hard to collect. It serves as the reference method in the empirical analysis.

PMT targeting food (in)security (PMT-FCS, T2). Beneficiaries are identified through an econometric PMT model designed to identify households with lower food security, as measured through the Food Consumption Score (FCS). The FCS is built using nine survey questions on the frequency of food groups consumed by the household in the past 7 days, and ranges from 0 to 112. This outcome might be considered the most obvious choice for food assistance programs, given that it targets food-related vulnerability.

PMT targeting a multidimensional vulnerability index (PMT-CARI, T3). Beneficiaries are identified through an PMT model designed to identify households with a combination of low expenditure per capita, low food security, and low capacity to cope with shocks (as captured through a set of 16 survey questions of coping related to assets). The CARI assigns households to four levels: food secure, marginally food secure, moderately food insecure, and severely food insecure. The two insecure categories are collapsed into a binary indicator, and the model predicts the probability of food insecurity. This method combines the economic dimension of T1 and the food-consumption dimension of T2, capturing both permanent and transient vulnerability. It is applied less often because the four-category indicator is harder to model.

Expert selection (Experts, T4). Beneficiaries are identified using a score based on priority dimensions identified through a survey administered to 28 local experts and practitioners from Lebanon. The experts were asked to indicate, in order of importance, their top-10 criteria for identifying and selecting Lebanese households in need of food assistance. The list of potential criteria mirrored the variables available for the PMT models. The answers were then combined and weights were attributed to each criterion, based on their frequency and ranking order. Each household was then assigned a score based on these weights.

3 Study Design

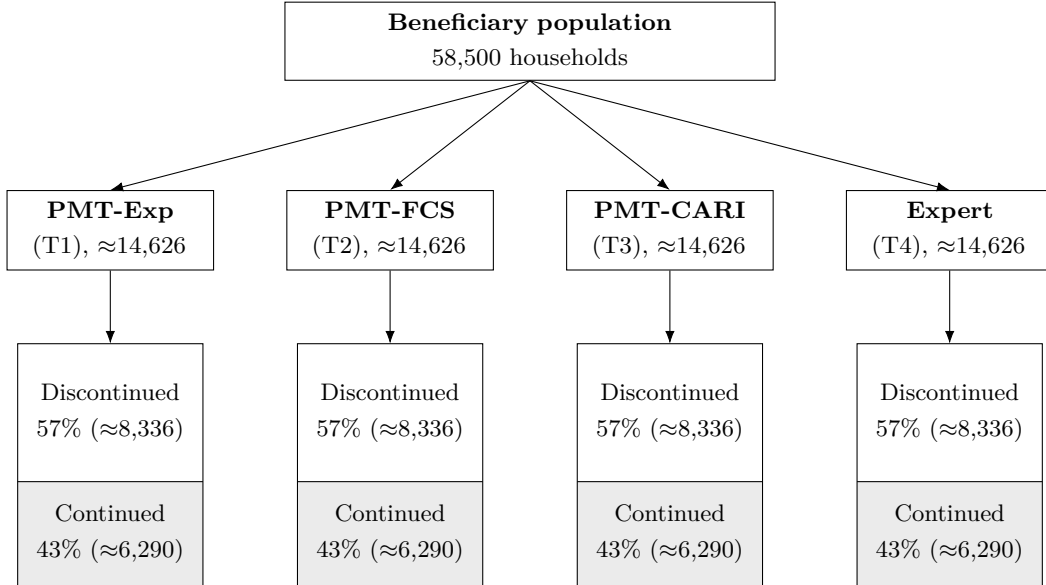
3.1 Experimental Design

Our study primarily builds on the individual-level randomization implemented by [Baliki et al. \(2025\)](#), in which the universe of beneficiaries (58,501 households) were randomly assigned to one of four targeting methods (T_j). We exploit this design in three ways. First, we compare

the performance of the four targeting approaches. Second, we combine the random assignment with the escalation of conflict between survey rounds to examine how targeting and assistance shaped households' responses to a major shock. Third, we exploit the variation in marginal eligibility generated by the four targeting methods to construct and evaluate an additional, impact-based targeting approach, and simulate its performance relative to the original four methods in protecting households against the conflict shock. We describe each analysis in turn below.

The comparison of targeting approaches relies on an individual-level randomization that allocated 58,501 assistance recipients across four equally sized targeting arms. Within each arm, households were ranked according to the assigned targeting method, and the 43% considered most vulnerable were selected for continued assistance, while support was discontinued for the remainder. This process retained 25,160 beneficiaries, approximately 6,290 per arm (out of 14,626), and discontinued assistance for the remaining 33,341.¹² All four methods are data-driven and use the same census data, so each can be replicated for every household. Prior to randomization, households were therefore classified according to their eligibility under each method, creating 16 mutually exclusive eligibility strata (2^4). Randomization was stratified by these strata to ensure balance across the different patterns of eligibility. Figure 1 illustrates the allocation across the four targeting arms.

Figure 1: Random allocation of households into the four targeting arms



Notes: The universe of 58,500 beneficiary households was randomly assigned, with stratification by eligibility space, to one of four targeting methods. Each method applies the same prioritization rule, retaining the 43% most vulnerable households (Continued) and discontinuing the remainder. Counts are approximate and equal across arms by construction.

The experiment compares the consequences of assistance discontinuation under alternative targeting approaches. Although all approaches applied the same retention threshold, they

¹²Most excluded households stopped receiving assistance in January 2024. A randomly selected subset of 960 households received an additional six months of support, until June 2024. This subgroup is used by Baliki et al. (2025) to study the effects of longer warning periods.

differed in how vulnerability was defined and measured, generating substantial variation in the households they selected. Our identification strategy exploits this variation in beneficiary assignment, as overlap across targeting methods is limited. The pairwise correlations in selection average 0.33 across the whole population. The three PMT-based methods are more alike, but their average correlation reaches only 0.55, and the Expert method correlates least with the others, from 0.02 with PMT-Exp to 0.18 with PMT-FCS. This imperfect overlap generates the cross-method variation on which our identification rests. Households that are eligible under all four methods (13%) or never eligible (26%) contribute no useful variation. Appendix Table A2 reports the full correlation structure, and Appendix Figure A2 shows the PMT-Expenditure score for selected households under each approach.

The discontinuation protocol remained intact through the first round of data collection. By the second round, however, the escalation of conflict and WFP’s emergency scale-up in heavily affected districts had partially overridden the original assignment. Consequently, 74% of households designated to stop receiving food assistance had instead received three months of emergency support. Discontinuation no longer matched the original design. We therefore restrict the discontinuation analysis to the first round, when the protocol still held, and use it to estimate how the targeting rule shaped its impact.

To examine whether targeting and continued assistance shaped how households weathered the conflict, we combine the escalation that occurred between survey rounds with the experiment’s random assignment to targeting methods. This analysis is motivated by the evidence that households experience and respond to conflict in highly heterogeneous ways, and that program effectiveness varies with conflict intensity (Stojetz et al., 2024; Brück et al., 2016).

Our empirical strategy exploits two sources of variation. The first is geographic variation in conflict exposure. We measure exposure at the cadaster level, the finest geographic unit available, using ACLED conflict events and fatalities recorded between survey rounds. Appendix Figure A4 shows that conflict remained low through the first survey round, surged after the October 2024 Israeli ground invasion between rounds, and stayed elevated thereafter. Following Moya (2018), we construct a continuous measure of conflict intensity and estimate its effects using a difference-in-differences framework that compares changes in household outcomes before and after the escalation across areas exposed to different conflict intensities. Identification relies on a parallel-trends assumption, which we assess using the pre-escalation survey rounds.

The second source of variation comes from the experiment’s random assignment to targeting methods. Although conflict exposure was concentrated in the south (see Appendix Table A6), it was unrelated to prior assistance receipt, and assignment to targeting methods was orthogonal to exposure. Combining this experimental variation with the panel data allows us to test whether continued assistance mitigated the effects of conflict and whether the targeting approaches differed in identifying households that were most affected by the shock.

We examine first whether continued assistance was associated with smaller conflict-related losses. Since assistance receipt depended on eligibility under the assigned targeting method, selection into continued assistance remains a consideration when interpreting these estimates. Reassuringly, baseline eligibility is unrelated to subsequent conflict intensity. Nevertheless, as a robustness check, we exploit the marginal-eligibility design of Baliki et al. (2025), which isolates

quasi-random variation in assistance receipt among households that are eligible under some targeting methods but not the one to which they were randomly assigned. We implement this design using an instrumental-variables specification that instruments actual assistance receipt with assignment-induced eligibility.

We then examine whether the effects of conflict differed across the four targeting arms by interacting conflict intensity with the assigned targeting method. The panel structure absorbs pre-conflict differences in outcomes, while random assignment ensures that the targeting method is orthogonal to conflict exposure. The interaction coefficients therefore capture whether households selected under a particular method experienced smaller conflict-related losses relative to the reference approach.

The final feature we exploit builds on the marginal-eligibility design of [Baliki et al. \(2025\)](#), which exploits the random assignment of households to targeting methods to isolate random variation in assistance receipt among households that are eligible under some targeting methods but not others. While [Baliki et al. \(2025\)](#) use this variation to estimate the effects of assistance discontinuation, we extend their approach by estimating household-level treatment-effect heterogeneity. The approach is close in spirit to [Haushofer et al. \(2025\)](#), who contrast targeting on deprivation (selecting the poorest households) with targeting on impact (selecting households with the largest predicted benefit from a transfer). In our setting, the treatment of interest is assistance withdrawal rather than assistance introduction.

Impact-based arm (Impact, T5). The impact-based targeting arm ranks households according to their predicted welfare loss from assistance discontinuation, measured by their household-level conditional average treatment effects (CATEs) at the first round of data collection. We estimate CATEs following the heterogeneous treatment-effect framework of [Chernozhukov et al. \(2018\)](#) for expenditure outcomes (total and food expenditure) and food-consumption outcomes (food consumption score and CARI). We find substantial treatment-effect heterogeneity only for expenditure outcomes. The estimated discontinuation effect on total expenditure is \$47.3 for households in the highest-impact tertile, compared with \$5.6 in the lowest tertile (a difference of \$41.7, SE 13.6, $p < 0.01$), with a comparable difference for food expenditure (\$16.1, SE 5.2, $p < 0.01$). Consistent with these results, the generic machine-learning calibration test rejects the null of treatment-effect homogeneity for both total expenditure (differential-forest coefficient 0.43, $p = 0.02$) and food expenditure (0.53, $p < 0.01$), whereas it cannot reject homogeneity for the food-consumption outcomes. Appendix Figure A5 presents the group average treatment effects (GATES) by tertile of the predicted effect.

We then sort households into groups by their predicted effect (CATE). We bring these predictions back to the census data to identify and rank households at baseline by their predicted welfare loss from discontinuation. The impact-based arm (T5) applies the same prioritization rule as the other methods, retaining the 43% of households with the largest predicted loss. The four vulnerability-based methods (T1–T4) are deprivation-based in this sense, whereas T5 operationalizes an impact-based alternative, calibrated with the effects of discontinuation at the first round of data collection.

While expenditure responses to assistance discontinuation are theoretically ambiguous, in the

context of an in-kind assistance program we interpret higher expenditure as an adverse response. Households are unlikely to reduce food consumption immediately and instead compensate for the lost transfer by purchasing food, often at the expense of other expenditures or through negative coping strategies (Martinchek et al., 2025). Consistent with this interpretation, Baliki et al. (2025) show that the increase in food expenditure after assistance withdrawal was financed through debt.

Having constructed the impact-based arm (T5), we compare the four deprivation-based rules (T1–T4) and the impact-based arm using two approaches. First, we simulate the welfare that each rule would deliver by continuing assistance to the 43% of households it selects, valuing household gains through a utility function that places greater weight on poorer households and expressing the result as the share of the welfare achieved under perfect targeting. Second, restricting the sample to the households selected by each rule, we compare the effects of the targeting methods on food security, coping, assets, and mental well-being.

3.2 Sampling

The main analysis is based on a sample of 6,400 households. The sample was constructed from 64 experimental cells, which are the strata of the randomization itself: households were stratified into the 16 eligibility groups within each of the four study arms T_j , producing $16 \times 4 = 64$ strata that serve as the sampling cells. From each cell, 100 households were sampled, aiming for a balanced number of eligible and non-eligible households, and therefore a balanced number of households that continue and that discontinue receiving assistance.

This balanced design does not reflect the actual population distribution across eligibility status and cells. Baliki et al. (2025) adopted it to maximize statistical power for measuring the impact of assistance discontinuation, which they identify by comparing eligible and non-eligible households. However, the eligibility status of every household in the full population is known, and the 100 households were sampled at random. Therefore, we can recover population-representative estimates by weighting observations. With 1,600 households per study arm, or 6,400 in total, the study can detect a difference across targeting methods of 0.1 standard deviations or larger on the primary outcome, at the 5% significance level with 80% power and assuming 10% attrition.

Appendix Table A3 reports how the sample is distributed across Lebanon’s eight governorates, alongside the corresponding shares in the full beneficiary population (58,501 households). Despite being drawn by eligibility cell rather than by location, the sample tracks the initial geographic distribution closely: Mount Lebanon is the largest governorate in both (28.5% of the sample versus 28.9% of the population), and the North, South, Beirut, Bekaa, and Akkar shares each differ by less than one percentage point. The two visible departures are El Nabatieh, which is somewhat underrepresented in the sample (7.8% versus 10.6%), and Baalbek-El Hermel, which is slightly overrepresented (9.4% versus 7.2%).

3.3 Data and Timeline

We combine four main sources of data for this study. First, we draw on the baseline beneficiary census of 58,501 households. Second, to train the PMT models, we make use of the Lebanon Household Survey (LHS), an external dataset covering a nationally representative sample. Third,

we use WFP administrative records, including information on program implementation. Finally, we rely on two rounds of household survey data collected from the stratified representative sample of 6,400 households. Data sources in more detail are as follows.

Beneficiary census: In preparation for its prioritization exercise, WFP conducted a census between August and September 2023 covering all 58,501 households included in the re-targeting activity. The census provides a detailed picture of beneficiary circumstances and serves two key purposes within this study. It offers the most up-to-date information for implementing the targeting exercise. It also establishes the baseline for the study sample. We further use an earlier version of this census, conducted in 2020-2021 and covering 7,400 households, to provide evidence on parallel trends among households that kept receiving assistance across census rounds and the impact evaluation.

Lebanon Household Survey (LHS): The LHS is a collaborative initiative between the World Bank, the United Nations High Commissioner for Refugees (UNHCR), and WFP, with support from the Joint Data Center on Forced Displacement, to monitor the socioeconomic conditions of Lebanese citizens and refugees in Lebanon. It is a nationally representative panel of about 3,500 households and records poverty, food security, employment, access to essential services, and coping mechanisms. One LHS round was collected between January and May 2023, a few months before the census. We use it to train the three PMT models on the 126 variables¹³ that overlap with the beneficiary census dataset.

Administrative data: The WFP Lebanon country office keeps detailed administrative records of its operations, collected independently of our study and following standard WFP procedures. Two of these records enter our analysis. To check adherence to the study protocol, we use monthly data on which households collected their food parcels. To gauge how beneficiaries received the exclusion decision, we use records of appeals filed by phone or through an online form. The country office also shared two further inputs: information on priority areas affected by the conflict with Israel, and records of households that received emergency cash transfers.

Household survey data: Our main analysis draws on the two rounds of survey data collected from the study sample of 6,400 households.¹⁴ The research team developed the questionnaires to cover a broad range of household characteristics and welfare measures, which we detail in the following section. The Central Arab Statistical Network (CASN), an independent survey firm with extensive experience in large-scale data collection across Lebanon, carried out the fieldwork. CASN deployed over 60 enumerators across the country for both rounds. The survey teams were familiar with local customs and were masked to the assignment status of the households they interviewed. All surveys were conducted in person using SurveyCTO, a secure data collection tool built on Open Data Kit (ODK).

¹³Standardizing these variables and adding interaction and squared terms yields 192 candidate machine-readable covariates (the candidate set summarized in Appendix Table A4); LASSO retains a sparser subset of these.

¹⁴The study was registered with the AEA RCT Registry (11831) and received ethical approval from the Humboldt University Ethics Committee (2023-08-W) and the HML International Review Board (2766).

Appendix Figure A6 shows the detailed timeline of the study, including the census data collection (August–October 2023), the two survey rounds (May–June 2024 and January–February 2025), and the timing of WFP’s in-kind transfers. Between the two rounds of data collection, the conflict between Lebanon and Israel along the southern Lebanese border escalated, leading to intense bombing across the country, population displacement, and a ground invasion in early October 2024. In October 2024, WFP started providing emergency cash transfers to households in affected areas, defined by conflict and displacement intensity, that were not already supported by its food assistance program. The second round of data collection thus took place while this emergency cash support was ongoing.

3.4 Metrics and Outcomes

Modeling performance

We start by studying how the targeting methods are modeled under each outcome, extending the work of Altındağ et al. (2021) on refugees. We benchmark a traditional Ordinary Least Squares (OLS) proxy-means test against several regularisation techniques: Ridge regression, the Least Absolute Shrinkage and Selection Operator (LASSO), and Elastic Net. We also include the generalized random forest approach recently applied by Haushofer et al. (2025). To capture three dimensions of deprivation (consumption poverty, food insecurity, and multidimensional food insecurity), we test alternative indicators and specifications for each dimension, selecting models that minimize the mean squared residuals (equivalently, maximize the out-of-sample R^2) under cross-validation. We then examine the predictors selected by the preferred models and quantify the contribution of different groups of household characteristics to understand which dimensions drive predictive performance.

Modeling metrics

- **Out-of-sample R^2 :** The share of outcome variance explained by the model’s predictions in held-out data not used for estimation. Unlike the in-sample R^2 , it rewards genuine predictive accuracy rather than in-sample fit and can be negative when predictions perform worse than the sample mean; it is the relevant criterion when the goal is prediction rather than explanation (Mullainathan and Spiess, 2017).
- **Predictor contribution:** Predictor importance is measured as the absolute value of the coefficient multiplied by the predictor’s standard deviation. Predictor contributions are normalized to sum to 100 percent within each model and then aggregated by predictor group, allowing comparisons across models.

Targeting performance

We assess the targeting alternatives along two complementary sets of measures: *performance measures*, which capture how accurately each method identifies deprived households, and *welfare outcomes*, which capture the household conditions that targeting seeks to improve and against

which its downstream effects are evaluated. The welfare outcomes are specified in the pre-registration.¹⁵ Except for formal complaints, obtained from WFP administrative data, all welfare outcomes are based on the household survey. Below, we describe each set of measures and the corresponding indicators in detail.

We evaluate *accuracy* of the predictions using *threshold-agnostic* measures that capture how well a method orders households by need, irrespective of any cutoff, and *threshold-dependent* measures that capture how well it classifies households at the eligibility cutoff. The census did not collect the outcomes used for targeting, so we assess predictive fit in the LHS, where we observe these outcomes.

Accuracy — threshold-agnostic measures

- **Spearman rank correlation:** Our targeting rule depends only on the ordering of households by need, as in most settings where organizations prioritize according to funding availability. We therefore follow Aiken et al. (2022) in preferring the Spearman correlation, which measures the agreement between the predicted and actual rankings of household welfare.
- **ROC curve and AUC:** Tracing the true- and false-positive rates across all candidate cut-offs yields the receiver operating characteristic (ROC) curve, proposed as a targeting-performance tool by Wodon (1997). The associated area under the curve (AUC) summarizes targeting quality in a single number, equal to 0.5 under random targeting and 1 under perfect targeting (McBride and Nichols, 2018; Aiken et al., 2022; Verme and Gagliarano, 2019).

Accuracy — threshold-dependent measures (at the 43% cut-off)

- **Inclusion and exclusion errors:** The inclusion error is the share of beneficiaries who are not deprived (type I), and the exclusion error is the share of deprived households left out (type II) (Coady et al., 2004; Hanna and Olken, 2018; Schnitzer and Stoeffler, 2024).
- **Precision and recall.** The share of selected households that are genuinely deprived, equal to $1 - \text{inclusion error}$, and the share of deprived households that are reached, equal to $1 - \text{exclusion error}$ (Aiken et al., 2022).
- **Coady–Grosh–Hoddinott (CGH) index:** The share of assistance received by the deprived divided by their share in the population, capturing the extra benefits the deprived obtain relative to a random or universal allocation. It equals 1 under random targeting and exceeds 1 when targeting is progressive (Coady et al., 2004; Schnitzer and Stoeffler, 2024).
- **Targeting differential:** Following Galasso and Ravallion (2005), the difference between the share of the deprived and the share of the non-deprived reached by a method, capturing

¹⁵Registered at the American Economic Association’s registry for randomized controlled trials (AEARCTR-0011831).

its progressivity. It ranges from -1 to 1 and equals 0 when a method reaches the deprived and non-deprived in equal proportion (Schnitzer and Stoeffler, 2024).

Welfare outcomes

Although the census did not collect the outcomes used for targeting, it allows us to construct an *asset index*, a common proxy for socioeconomic status in the absence of consumption or expenditure data (Filmer and Pritchett, 2001; Sahn and Stifel, 2003). We use it to assess the progressivity of each targeting method by comparing the asset distribution of selected households with that of the full census population. The asset index does not capture transient needs such as food insecurity (Banerjee and Duflo, 2007). It is nevertheless the only welfare proxy available across the full census, so it anchors our assessment of beneficiary selection.

- **Asset index.** We extract the first principal component from a principal component analysis (PCA) of 33 binary indicators of household asset ownership—ranging from basic durables such as a refrigerator, washing machine, and water heater to communication and higher-value items such as a flat-screen television, satellite dish, smartphone, computer, air conditioner, and car—and standardize it so that higher values indicate greater wealth (Filmer and Pritchett, 2001; Vyas and Kumaranayake, 2006; McKenzie, 2005).

The household survey collected a broad set of welfare outcomes spanning food security, coping behavior, consumption, displacement, and psychological well-being:

- **Food security:** The Food Consumption Score (FCS), which captures dietary diversity and frequency of consumption across eight food groups over a seven-day recall period, producing a score between 0 and 112.
- **Reduced Coping Strategies Index (rCSI):** Captures the frequency and severity of five short-term, food-related coping behaviors over the past seven days, such as eating less-preferred foods, limiting portion sizes, or reducing the number of meals (Maxwell et al., 2008; Maxwell and Caldwell, 2008).
- **Livelihood Coping Strategies Index (LCSI):** Records whether households resorted to longer-term, livelihood-eroding strategies—stress, crisis, and emergency coping such as selling assets or cutting education and health expenditures—following WFP’s standard food-security methodology (WFP, 2021).
- **Expenditure:** Measured per capita and per month, following WFP’s standard methodology, in two components:
 - *Food expenditure:* is the first outcome we consider to understand how the drop in food assistance reshaped households’ decisions. Households reported expenditure across nine food groups using a standard 30-day recall period, and the survey captured not only actual disbursements but also food purchased on credit and the estimated value, at current prices, of own food production. The outcome is winsorized at the 95th percentile.

- *Non-food expenditure*: is a non-exhaustive list of non-food expenses covering the most common sources of spending among the target population, using two recall periods: 30 days for 11 regular expenses such as electricity and transport, and six months for 7 less frequent expenses such as health care or asset purchases, the latter divided by six to obtain a monthly estimate. The outcome is winsorized at the 95th percentile.
- **Displacement**: Whether the household was displaced during the conflict, defined as having left its habitual residence since the July 2024 escalation. We also record whether the move was forced or voluntary, whether the household remained displaced at the time of the survey, how long it stayed away, and the type of shelter it used while displaced.
- **Psychological well-being**: Captured through complementary indicators:
 - *WHO-5 mental health index*: Assesses psychological well-being over the past two weeks using five items on cheerfulness, calmness, rest, energy, and interest in daily life, rescaled to a 0–100 scale. We combine these items into a standardized inverse covariance-weighted index following [Anderson \(2008\)](#).
 - *Trauma index*: Severity of post-traumatic stress symptoms, measured with the PTSD module of the International Trauma Questionnaire (ITQ), a self-report measure of ICD-11 PTSD ([Cloitre et al., 2018](#)). Respondents report how much each of six symptoms bothered them in the past month, two for each of re-experiencing, avoidance, and sense of current threat, which we sum into a severity score; three further items record functional impairment in relationships, work, and other important areas. We combine these items into a standardized inverse covariance-weighted index following [Anderson \(2008\)](#).

A detailed table with the construction of indexes and other indicators can be found in Appendix Table [A5](#).

4 Empirical Strategy and Internal Validity

4.1 Empirical Strategy

Targeting performance

The modeling performance and accuracy of the targeting models are evaluated using standard metrics from the data-driven targeting literature, following the evaluation framework employed by [Altındağ et al. \(2021\)](#), [Aiken et al. \(2022\)](#), and [Schnitzer and Stoeffler \(2024\)](#).

To assess differences in satisfaction, measured through complaints, and the downstream welfare implications of alternative targeting methods, we exploit the randomized assignment of households. Using the first round of data collection, we compare follow-up outcomes across experimental arms. We estimate intent-to-treat (ITT) effects with the following regression specification:

$$Y_{he} = \beta_1 \text{Targ}_{he}^2 + \beta_2 \text{Targ}_{he}^3 + \beta_3 \text{Targ}_{he}^4 + \delta_e + \Omega X_{he} + \varepsilon_{he} \quad (1)$$

where Y_{he} is an outcome of interest (e.g., whether the household submitted a complaint, the household’s FCS, etc.) for household h in eligibility space e . Targ_h^i indicates assignment to targeting method i , with method 1 (PMT-Exp) as the reference category throughout. The specification includes eligibility-space fixed effects δ_e , corresponding to our level of stratification, and X_{he} , a vector of baseline control variables selected through a double-lasso procedure, which always includes a control for households randomly assigned to the long notice window. Finally, ε_{he} is the error term, and robust standard errors are reported throughout.

The coefficients β_i capture the average difference in Y between households assigned to targeting method i and the reference targeting method. By comparing the coefficients among each other, we can also assess the relative performance of the targeting methods against each other. When focusing on complaints, the administrative data cover the entire population of 58,501 households. For all other outcomes, instead, the analysis is based on survey data collected from the study sample of 6,400 households.

Our main specification reports unweighted estimates for simplicity. However, since the study oversamples smaller eligibility spaces, the appendix reports estimates weighted to recover population-average effects. The results are nearly identical, indicating that weighting does not affect our conclusions.

Equation (1) estimates the average impact of each targeting approach on the full population randomized into each arm, including both beneficiaries and non-beneficiaries. To examine whether effects differ across groups, we re-estimate Equation (1) separately within subgroups of interest, rather than interacting the arm indicators with covariates. Specifically, we estimate the model separately for beneficiaries and non-beneficiaries, as well as for subgroups defined by baseline household characteristics. For a subgroup g , we restrict the sample to its members and estimate

$$Y_{he} = \beta_1^g \text{Targ}_{he}^2 + \beta_2^g \text{Targ}_{he}^3 + \beta_3^g \text{Targ}_{he}^4 + \delta_e + \Omega X_{he} + \varepsilon_{he}, \quad (2)$$

where β_i^g represents the effect of targeting method i relative to the reference method within subgroup g . For subgroups defined by baseline characteristics, differences in β_i^g describe how treatment effects vary across ex ante household characteristics. The beneficiary/non-beneficiary comparison should instead be interpreted descriptively, as beneficiary status is determined by the targeting method itself and is therefore not randomly assigned.

Conflict exposure and the role of targeting

Households are observed repeatedly over time, and the conflict escalation occurred between the two follow-up survey rounds. We exploit the panel structure of the data and geographic variation in conflict exposure to implement a difference-in-differences design, allowing us to estimate the impact of the escalation and the role of targeting in moderating it (Moya, 2018). Specifically, for household h observed in round t , we estimate the following difference-in-differences specification, relating changes in household welfare over time to geographic variation in conflict exposure:

$$Y_{ht} = \alpha_h + \lambda_t + \beta (\text{Post}_t \times \text{Exp}_h) + \varepsilon_{ht}, \quad (3)$$

where Y_{ht} is a welfare outcome, α_h a household fixed effect, λ_t a round fixed effect, Post_t indicates the post-escalation (Round 2) round, and Exp_h is a continuous measure of conflict intensity in household h 's locality (locality-level ACLED activity between the rounds). The coefficient β is the effect of the conflict per unit of exposure—the differential change in welfare for more- versus less-exposed households. Standard errors are clustered at the locality level.

Identification rests on a parallel-trends assumption: absent the escalation, more- and less-exposed households would have followed parallel welfare trajectories. We assess it with the pre-escalation rounds, testing whether exposure predicts differential changes in welfare before the conflict, and restrict attention to outcomes whose pre-conflict trends do not differ by exposure.

To ask whether the targeting method shaped how households weathered the conflict, we interact the conflict treatment with the randomly assigned targeting arm, which turns Equation (3) into a triple difference:

$$Y_{ht} = \alpha_h + \lambda_t + \beta (\text{Post}_t \times \text{Exp}_h) + \sum_i \pi_i (\text{Post}_t \times \text{Targ}_h^i) + \sum_i \delta_i (\text{Post}_t \times \text{Exp}_h \times \text{Targ}_h^i) + \varepsilon_{ht}. \quad (4)$$

The household fixed effects absorb the time-invariant arm and exposure terms and their interaction, and the round fixed effects absorb Post_t . The coefficients of interest are the triple interactions δ_i , which capture whether the conflict's effect on welfare, per unit of exposure, differs for households assigned to targeting method i relative to the reference method. Since assignment to a targeting arm is random and orthogonal to conflict exposure, δ_i identifies the moderating role of the targeting method itself.

Welfare comparison of deprivation- and impact-based targeting

Constructing the impact-based arm requires household-level estimates of the welfare loss from assistance withdrawal. We estimate the conditional average treatment effect (CATE) of discontinuation, $\tau(x) = \mathbb{E}[Y_h^{\text{cut}} - Y_h^{\text{cont}} \mid X_h = x]$, using a causal forest (Wager and Athey, 2018; Athey et al., 2019) that exploits the randomized variation generated by the targeting assignment and the marginal-eligibility design. Following Chernozhukov et al. (2018), the forest is estimated using honest trees and out-of-bag predictions, so each household's treatment effect is predicted using only trees that exclude that household. We then rank households by their predicted welfare loss, $-\hat{\tau}_h$, and retain the top 43%, separately for total and food expenditure per capita. These predicted losses also provide the welfare gains, $g_i = -\hat{\tau}_i$, used in the welfare simulations.

To compare targeting rules on a common scale, we simulate the social welfare each would deliver under a fixed budget, following Aiken et al. (2025) and Hanna and Olken (2018). For targeting method m , which selects the set of households S_m (its prioritized 43%), each household's simulated outcome equals its outcome under assistance withdrawal, y_i^{cut} , plus the benefit of continued assistance, g_i , if it is selected ($i \in S_m$). Social welfare is then

$$V_m = \sum_i \omega_i U(y_i^m), \quad (5)$$

where ω_i are survey-by-household-size weights. We use a CRRA utility function, $U(c) =$

$(c^{1-\rho} - 1)/(1 - \rho)$, for per-capita expenditure, setting the coefficient of relative risk aversion—equivalently, the planner’s aversion to inequality—to $\rho = 3$. This value places substantial but not extreme weight on welfare gains to poorer households and is the standard choice in the targeting and social-welfare literature (Hanna and Olken, 2018; Aiken et al., 2022, 2025).

Following Aiken et al. (2025), we summarize each method by its welfare-capture ratio

$$G_m = \frac{V_m - V_0}{V_{1B} - V_0}, \quad (6)$$

which measures the share of the attainable welfare gain captured by the targeting rule. Here, V_0 denotes welfare if assistance is withdrawn from all households, while V_{1B} is the first-best allocation that selects the same number of households but those with the largest utility gains from continued assistance. Thus, $G_m = 1$ corresponds to perfect targeting, while random allocation provides the no-information benchmark. Inference is based on bootstrap resampling within eligibility strata.

The impact-based rule is constructed to maximize welfare on expenditure. To assess whether its advantage extends beyond the outcome it targets, we repeat the same welfare-simulation exercise for food security, coping, durable assets, and mental well-being, using the same CRRA welfare framework ($\rho = 3$). This allows us to assess whether ranking households by their predicted expenditure losses also improves welfare along other dimensions, or whether its advantage is specific to expenditure.

4.2 Internal Validity

Balance tests confirm that randomization successfully generated four comparable groups. Appendix Table A7 shows the results of regressing study arm assignment on 153 baseline variables, with no strong evidence of systematic differences. As expected under random assignment, only a small proportion of tests show statistically significant differences: 7.8% of p-values are below 0.10, 5.2% are below 0.05, and only 1.3% are below 0.01. These rates align with expectations under the null hypothesis of no difference, which implies a uniform distribution of p-values between zero and one (Bruhn and McKenzie, 2009). For example, approximately 5% of tests are expected to yield significance at the 5% level purely by chance.

Attrition is low and balanced across targeting arms. Excluding the long-window subsample discussed below, we surveyed 99% of the target sample at the first round and 94% at the second round, with no differential attrition across the four arms (F-tests yield $p > 0.1$ in both rounds) or in the corresponding regressions (reported in the separate supplementary analysis). When we could not reach a target household, we replaced it with another from the same cell; the main analysis uses the full sample, and all results are robust to excluding replacements. First-round coverage was lower for the long-window households, which belong to a separate randomization in Baliki et al. (2025) that we do not use in this study.¹⁶ We corrected the error before the second round, when coverage rose to 93% for this subsample. Attrition is likewise balanced across the discontinuation and long-notice randomizations (not part of this study).

¹⁶The lower coverage came from a coding error: when drawing the survey sample, the code re-randomized the long-window households without accounting for their existing assignment under that earlier randomization. We surveyed 31% of them in the first round, still randomly selected from the discontinued households.

In terms of adherence to the study protocol, we verified this through the detailed administrative records maintained by WFP. As part of its standard procedures, WFP tracks each parcel and beneficiary, enabling the monitoring of assistance receipt on a monthly basis. Administrative data confirm that by the time of data collection (June 2024) 97% of households expected to continue receiving assistance in 2024 did so, while 99% of households expected to stop receiving assistance did not receive it. Survey data also collected self-reported information on assistance receipt. Although these data appear noisier, potentially due to recall errors or (strategic) misreporting, they align very well with the expected patterns.

The conflict analysis relies on three conditions, which we assess in turn. The first is that conflict exposure is orthogonal to the randomly assigned targeting arm, so that the triple-difference coefficients δ_i in Equation (4) isolate the moderating role of the targeting method. Appendix Table A8 supports this: the share of households in an exposed locality, and the mean number of conflict events and fatalities, are statistically indistinguishable across the four arms (F-test $p = 0.23, 0.61$, and 0.77). Conflict exposure is also unrelated to households' prior receipt of assistance (see Appendix Table A10). Continued and discontinued households were equally likely to be exposed (56% in each group) and experienced identical mean conflict intensity (0.93 versus 0.93, $p = 0.96$). The continued-versus-discontinued comparison in Section 5.4 is therefore not confounded by differential exposure. Conflict was, however, far from randomly distributed across households: more exposed localities were poorer and more likely to be female-headed or workless. Accordingly, identification relies on a difference-in-differences design with household fixed effects rather than cross-sectional variation.

WFP's emergency scale-up following the conflict escalation is unlikely to affect our displacement measure, as most displacement had already occurred before households began receiving emergency assistance. Other outcomes measured in the second survey round, including expenditure, food security, and mental well-being, may instead partly reflect the effects of these transfers, since many households were receiving emergency support by then.

The emergency scale-up is the only post-assignment characteristic that is not perfectly balanced across targeting arms. Emergency-cash receipt, realized after assignment, is modestly higher in the Expert arm (60% versus 54–57% in the other three; joint $p = 0.003$, see Appendix Table A8). The difference is small, and with the same emergency transfer reaching a majority of households in every arm, it is more likely to attenuate cross-arm differences than to generate them. Moreover, we observe emergency-cash receipt at the household level and control for it directly. The estimated differences across targeting arms are virtually unchanged whether or not this control is included.

The third condition is parallel pre-trends. We assess it over the two pre-conflict rounds that precede the escalation—an earlier version of the census collected in late 2020–2021 and the first round of data collection (May–June 2024). Targeting arms are balanced in the earlier round (see Appendix Table A9), and the exposure-by-wave coefficient is indistinguishable from zero in the pre-period for the reduced-coping and asset-sale outcomes (see Appendix Figure A7), consistent with a common trajectory absent the shock. The Food Consumption Score is the exception: exposure predicts a differential pre-conflict change in FCS in the earlier round ($p < 0.01$), likely reflecting changes in the food basket households received across the census data-collection period

rather than the subsequent conflict. We therefore exclude FCS from the conflict analysis and restrict attention to outcomes whose pre-conflict trends do not differ by exposure.

5 Results

5.1 Targeting Performance

Modeling and predictors

Appendix Table A4 compares the out-of-sample predictive performance of OLS, Ridge, Elastic Net, LASSO, and the Generalized Random Forest (GRF). The three PMTs using the regularized methods perform similarly and match or outperform both OLS and GRF across all outcomes. For expenditure, the regularized specifications and OLS achieve nearly identical accuracy (out-of-sample R^2 of 0.51–0.52), while the GRF trails ($R^2 = 0.44$). For the Food Consumption Score, LASSO achieves the highest predictive accuracy ($R^2 = 0.16$), narrowly outperforming OLS and the GRF ($R^2 = 0.15$). For food insecurity (CARI), the regularized logit models again lead ($D^2 = 0.26$ for LASSO), ahead of the GRF ($D^2 = 0.23$) and well above the linear probability model ($D^2 = 0.17$). These results indicate that the more flexible GRF offers no predictive gains over LASSO. This is consistent with Aiken et al. (2025) and Cao et al. (2025), who likewise find that simpler machine-learning approaches perform as well as or better than more flexible alternatives. A likely explanation is that our cross-sectional specifications already include interactions and squared terms among the 192 candidate covariates, leaving little scope for additional gains from GRF. LASSO therefore provides the best balance between parsimony and predictive accuracy and is used to construct the PMT-based targeting arms.

We next examine the predictors driving selection under each method. Appendix Table A12 reports the ten most important predictors for each method and Appendix Figure A8 displays their contributions side by side, while Appendix Figure A9 groups them into durable assets, demographic characteristics, and socioeconomic conditions.¹⁷ The three PMTs differ mainly in the relative importance they assign to these characteristics. PMT-Exp is driven primarily by household size and housing conditions, with household size, crowding, and area per member accounting for more than half of total predictor importance. PMT-FCS relies overwhelmingly on durable assets, with internet access, a vacuum cleaner, a car, and white goods together contributing more than 80%. PMT-CARI places substantial weight on both dimensions. By contrast, the Expert rule relies almost entirely on categorical vulnerability indicators, particularly head unemployment (20%), marital status, disability of the head or another household member, and household composition, while placing essentially no weight on assets.

The contribution profile also has implications for data-collection costs. Across all methods, predictor importance declines rapidly, with the top ten variables accounting for 80–86% of total model contribution (Appendix Figure A10). This suggests that a shorter questionnaire or a population registry could reproduce most of the targeting at lower cost. However, the most important predictors differ across outcomes, so no single reduced set of variables suits every targeting objective. In our setting, the Expert rule relies primarily on stable demographic

¹⁷We refit the LASSO on the census to relate the variables observed for every household to the final predictions. The same exercise could equivalently be performed on the LHS used to estimate the PMTs.

characteristics, making it well suited to a population registry. By contrast, the PMTs place greater weight on durable assets, which require periodic updating, making them less suitable for a registry-based approach.

Accuracy (training data validation)

Table 1 reports the predictive accuracy of each targeting method on the LHS training dataset, evaluated against the outcome it was designed to predict, with the full method-by-outcome matrix shown in Appendix Table A13. Precision and recall assess classification at the 43% eligibility threshold, while the Spearman rank correlation and the area under the curve (AUC) summarize performance across the full distribution.

Table 1: Targeting accuracy: each method against the outcome it predicts

Method	Spearman ρ	AUC	Precision	Recall	Incl. err.	Excl. err.
PMT-Exp	0.57 [0.51, 0.62]	0.83 [0.80, 0.86]	0.69 [0.64, 0.75]	0.72 [0.67, 0.77]	0.31	0.28
PMT-FCS	0.35 [0.28, 0.41]	0.71 [0.66, 0.74]	0.58 [0.52, 0.63]	0.62 [0.56, 0.68]	0.42	0.38
PMT-CARI	0.45 [0.39, 0.51]	0.77 [0.73, 0.80]	0.62 [0.56, 0.67]	0.69 [0.64, 0.75]	0.38	0.31
Random	0.00 [-0.08, 0.07]	0.50 [0.46, 0.54]	0.41 [0.34, 0.45]	0.43 [0.38, 0.47]	0.59	0.57

Notes: Held-out LHS fold; each PMT against its target outcome (PMT-Exp: expenditure; PMT-FCS: FCS; PMT-CARI: CARI). Spearman’s ρ and the AUC are threshold-agnostic; the remaining metrics are at the 43% cut-off. Inclusion error = 1 – precision (selected but non-deprived); exclusion error = 1 – recall (deprived but not selected). *Random* is the no-information benchmark; the Expert score, lacking a predicted outcome, appears only in Appendix Table A13. 95% bootstrap CIs (1,000 replications) in brackets.

Each PMT predicts its target outcome substantially better than random targeting (AUC = 0.50). PMT-Exp performs best, ranking households by expenditure with an AUC of 0.83 (95% CI [0.80, 0.86]), followed by PMT-CARI for food insecurity (0.77) and PMT-FCS for the Food Consumption Score (0.71). The full method-by-outcome matrix (Appendix Table A13) shows that each PMT performs best against its own target outcome and consistently outperforms the Expert rule, which records the lowest accuracy across all outcomes. Against expenditure, for example, the PMTs achieve AUCs between 0.77 and 0.83, compared with 0.67 (95% CI [0.63, 0.71]) for the Expert rule, with non-overlapping confidence intervals. The same ordering holds for progressivity: Appendix Table A14 shows that the PMTs achieve substantially higher Coady–Grosh–Hoddinott indices and targeting differentials than the Expert rule (e.g., a CGH index of 1.67 for PMT-Exp versus 1.32). Overall, the observed levels of targeting accuracy are in line with previous evidence from proxy-means tests (Coady et al., 2004; Altındağ et al., 2021).

Accuracy (beneficiary census validation)

While we do not observe the outcomes used for targeting or the downstream impact measures in the census, we assess the progressivity of each targeting approach using the asset index, a welfare proxy available for every household. Figure 2 plots the relationship between the asset

index and the probability of program selection. Appendix Table A15 reports the corresponding regression estimates. Across all targeting arms, poorer households were significantly more likely to be selected ($p < 0.01$ in every case). The estimated log-odds range from -1.08 to -2.02 . The strongest associations are observed under PMT-FCS and PMT-CARI (-2.02 and -1.76 , respectively), while the Expert method shows the weakest relationship (-1.08). These differences are statistically significant across targeting arms (joint test of equal slopes, $p < 0.01$), indicating that all methods prioritize poorer households, but PMT-FCS does so most strongly and the Expert method the least.

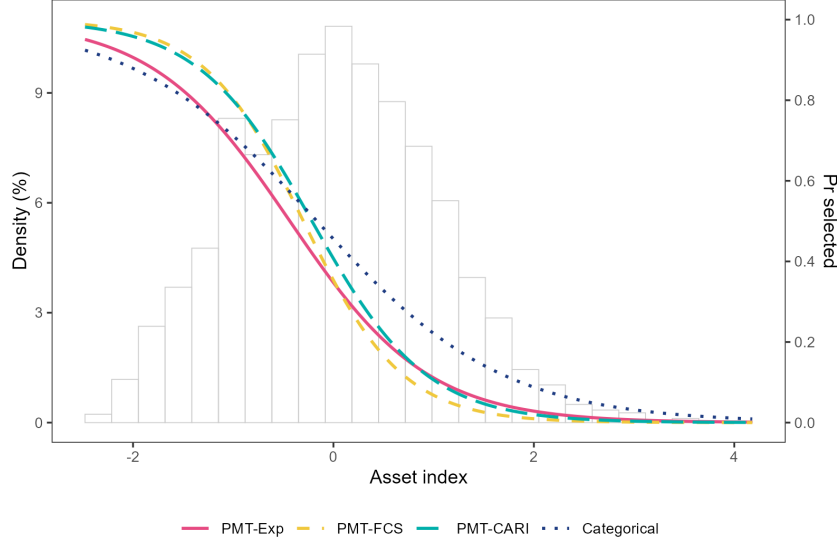


Figure 2: Progressiveness in selection using the asset index by targeting method

Notes: The figure plots the relationship between the baseline asset index and program selection across targeting methods. The background histogram (gray) shows the non-parametric probability distribution of the asset index. Overlaid are predicted selection probabilities from logit regressions, separately estimated for each targeting arm (PMT-Exp., PMT-FCS, PMT-CARI, and Expert). The curves illustrate the consistently negative association between household welfare and the probability of selection. See Appendix Table A15 for regression coefficients and further details.

5.2 Eligibility

Although all methods prioritize relatively disadvantaged households, they identify substantially different beneficiaries. Appendix Table A2 reports pairwise correlations in selection across the four targeting methods. The strongest overlap is between PMT-Exp and PMT-CARI (0.65), followed by PMT-CARI and PMT-FCS (0.59). PMT-Exp and PMT-FCS overlap more modestly (0.41). By contrast, the Expert method has little overlap with any PMT, with correlations ranging from 0.02 to 0.18.

The correlations summarize overlap but not whom each method selects. To characterize the selected households directly, Appendix Table A11 reports the mean of a broad set of baseline characteristics among the 43% of households retained under each method. We report the 20 characteristics on which the four methods differ most, ordered from the largest cross-method spread to the smallest. The largest differences concern household composition and the profile of the household head, and are driven almost entirely by the Expert method. The three

PMT methods select large households—4.0 to 4.7 members on average, above the population mean of 3.9, with more children and more working-age members—whereas the Expert method selects small households (3.2 members, 0.5 children under 15). The Expert method also selects households whose head is markedly older (62 vs. 52–55 years), far less likely to be married (41% vs. 66–76% under the PMTs), and far more likely to be elderly (46% vs. 17–25%), female (54% vs. 25–32%), or to have a disability (39% vs. 19–22%). Its households are also more often single-parent (40% vs. 16–21%) and far more likely to have no employed member (60% vs. 28–36%). The PMT methods prioritize larger, working-age households, whereas the Expert method prioritizes households with categorical markers of vulnerability, including old age, female headship, disability, single parenthood, and the absence of an employed breadwinner.

Where the methods converge is on economic deprivation. All four select households well below the population mean on the asset index (between -0.41 and -0.65 , versus a population mean of zero), confirming that every method is progressive despite prioritizing different dimensions of vulnerability. Differences in the head’s education are comparatively small. Variation among the three PMTs is also modest, although PMT-FCS tends to select somewhat older, more single-parent, and more asset-poor households than PMT-Exp or PMT-CARI. These differences in beneficiary profiles raise the question of whether the targeting methods also differ in the outcomes they deliver. We explore this question next.

5.3 Targeting Downstream Outcomes under Assistance Discontinuation

We begin with households’ own reaction to being discontinued, namely whether they lodged an appeal against the decision. Appeals are recorded in the administrative data for the full beneficiary census, so they can be compared across all four arms. Appendix Figure A3 shows the share of discontinued households that appealed, by targeting method. Appeal rates are similar under the three proxy-means tests (50–52%) but markedly higher under the Expert rule, at 56%, six percentage points above the PMT-Expenditure reference ($p < 0.01$). The Expert rule discontinues a distinct, less asset-poor population selected on categorical criteria, and its decisions appear less accepted by affected households, consistent with concerns about the perceived fairness of coarse categorical rules.

We then compare downstream welfare outcomes across the full study population, including both beneficiaries and non-beneficiaries. Table 2 shows no consistent differences across targeting arms. Relative to the PMT-Expenditure reference group, the estimated effects on food security (FCS), coping (rCSI), food and total expenditure, the three livelihood-coping margins (LCSI stress, crisis, and emergency), and mental well-being (WHO-5) are generally small and statistically insignificant. The handful of coefficients that reach significance are scattered and do not form a consistent pattern—PMT-FCS households, for instance, report somewhat higher food and total expenditure but show no advantage on FCS, coping, or well-being—and the tests of equality across the PMT and Expert arms rarely reject. Appendix Figure A11 likewise shows that the distributions of these outcomes are nearly identical across the four groups. Survey-weighted estimates of these downstream outcomes yield the same conclusion (Appendix Table A19).

Table 2: Impacts of Targeting on Downstream Outcomes (ITT)

	FCS	rCSI	Food Exp. (USD)	Total Exp. (USD)	LCSI Stress (pp)	LCSI Crisis (pp)	LCSI Emer- gency (pp)	WHO- 5 Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Reference method: PMT-Exp</i>								
PMT-FCS	0.16 (0.58)	-0.54 (0.43)	6.94** (2.83)	16.52** (7.47)	1.29 (0.80)	3.09* (1.75)	-0.25 (0.98)	0.37 (0.67)
PMT-CARI	1.06* (0.58)	0.34 (0.44)	2.27 (2.85)	6.73 (7.48)	0.56 (0.83)	1.33 (1.75)	0.44 (1.00)	0.12 (0.68)
Expert	-0.04 (0.58)	0.04 (0.44)	4.65* (2.80)	13.76* (7.53)	1.55** (0.79)	-0.58 (1.75)	-1.31 (0.95)	-0.08 (0.66)
Mean PMT-Exp.	54.47	16.62	136.64	392.21	93.97	56.07	8.42	23.00
Observations	6,358	6,358	6,358	6,358	6,358	6,358	6,358	6,300
p-val(PMT-FCS=PMT-CARI)	0.120	0.047	0.100	0.187	0.346	0.313	0.483	0.712
p-val(PMT-FCS=Expert)	0.739	0.186	0.412	0.711	0.731	0.036	0.261	0.498
p-val(PMT-CARI=Expert)	0.061	0.505	0.398	0.347	0.197	0.278	0.068	0.767

Notes: PMT-Expenditure targeting serves as the reference group. All regressions include randomization-strata (eligibility-cell) fixed effects. The table shows regression results on pre-specified primary outcomes at the household level using the wave of household-survey data described in Section 3. Each column is its own dependent variable. The independent variable of interest is the ITT treatment indicator and includes beneficiaries and non-beneficiaries. Food Consumption Score (FCS, 0–112) measures dietary diversity and frequency. The reduced Coping Strategy Index (rCSI, 0–56) measures frequency and severity of negative coping strategies. Expenditure includes credit purchases and own production but excludes program transfers; food and total expenditure are winsorized at the 95th percentile. The Livelihood Coping Strategy Index (LCSI) indicators equal one if the household adopted at least one stress-, crisis-, or emergency-level coping strategy, respectively; coefficients and means are in percentage points. Mental health is measured with the WHO-5 (0–100). The p -values at the bottom test the null hypothesis of equal treatment effects between targeting arms. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In summary, the results show that there is no clear targeting method that consistently outperforms the others. There are modest adverse effects on LCSI coping detected under the Expert approach, but they are small in magnitude. These findings are consistent with the limited evidence from experimental assessments of targeting, which generally shows that different approaches lead to the selection of different beneficiary groups, but such variation does not necessarily translate into differences in program effectiveness as measured by the average welfare of the targeted population. For instance, [Alatas et al. \(2012\)](#) find no significant differences in welfare outcomes when comparing proxy means testing (PMT) and community-based targeting (CBT). Similarly, [Premand and Schnitzer \(2021\)](#) find no differences in FCS or expenditure between households targeted through PMT versus CBT. However, they report that food stocks are on average 16% higher across all households in villages randomized to PMT compared to villages randomized to CBT, and 21% higher compared to villages assigned to FCS-based targeting. Still, it is possible that the observed average masks substantial heterogeneity across sub-groups.

Heterogeneous effects across subgroups

We next examine whether the targeting methods affect some households differently. Appendix Table [A17](#) reports, for several baseline subgroups, the effect of each method relative to PMT-

Expenditure on food security (FCS), coping (rCSI), food expenditure, and mental well-being (WHO-5). Estimates are reported separately for households within each subgroup. The subgroups are defined at baseline before randomization and include female-headed households, households with an elderly head (aged 65 or older), large households (five or more members), households in the bottom asset quartile, and households classified by eligibility under their assigned targeting method.

The clearest differences arise for households with an elderly head. Relative to PMT-Expenditure, they fare better under both PMT-FCS—food expenditure USD 18.6 higher ($p < 0.01$), FCS 2.2 points higher (4%, $p < 0.10$) and rCSI 1.3 points lower (8%, $p < 0.10$)—and the Expert rule—FCS 2.6 points higher (5%, $p < 0.05$), rCSI 1.7 points lower (11%, $p < 0.05$), and food expenditure USD 10.6 higher ($p < 0.10$). The poorest households show the same tilt: those in the bottom asset quartile do better under PMT-FCS, with food expenditure USD 15.6 higher ($p < 0.01$), rCSI 2.4 points lower (13%, $p < 0.01$), and WHO-5 well-being 2.5 points higher ($p < 0.10$). For female-headed households the differences are smaller and mixed—a 2.1-point higher FCS under PMT-CARI (4%, $p < 0.05$), with no consistent pattern elsewhere—and for large households no method differs significantly from PMT-Expenditure on any outcome.

On balance, the methods that more often select households with categorical vulnerabilities—PMT-FCS and the Expert rule—deliver modestly better outcomes for those groups than for the population as a whole. Consistent with this, among households that continued receiving assistance, outcomes are lower under PMT-FCS and the Expert rule than under PMT-Expenditure (under the Expert rule, food expenditure is USD 19.4 lower ($p < 0.01$) and WHO-5 well-being is 2.2 points lower ($p < 0.05$), with the full comparison by eligibility status in Appendix Table A18). Eligibility, however, is determined by each targeting method, so these comparisons jointly reflect differences in beneficiary selection and the effects of assistance discontinuation.

A final analysis focuses on the households over which the targeting methods actually disagree. Some households are selected by all four methods, some by none, and others only by a subset of methods. The first two groups receive the same treatment regardless of the targeting rule, so only the marginal households selected by some methods but not others can generate differences across targeting approaches.

Appendix Table A16 shows that the methods agree on the poorest households. Those selected by all four methods have the lowest baseline expenditure (USD 91 per capita, compared with USD 112–120 among households selected by none or only a few methods). Yet the welfare gains from continued assistance are remarkably similar across these groups, indicating that they are unrelated both to baseline deprivation and to the number of methods selecting a household. The households over which the methods disagree therefore generate welfare gains close to the sample average, so reallocating them across targeting rules has little effect on aggregate welfare. This is consistent with the evidence in Section 5.5, suggesting that disagreement at the eligibility margin reflects prediction error or transitory deprivation rather than persistent differences in need.

We test this directly by restricting the analysis to the marginal households on which the methods disagree. The full-population comparison in Table 2 includes households selected by all four methods and households selected by none, whose treatment status is identical regardless of

the targeting rule and therefore cannot contribute to differences across arms. Table 3 instead repeats the ITT analysis using only households selected by some, but not all, methods ($N = 5,566$, excluding the 404 always-eligible and 388 never-eligible households). The results are unchanged. The targeting arms remain statistically indistinguishable, the point estimates closely mirror those from the full sample, and only the same isolated differences remain significant, with PMT-FCS showing modestly higher food and total expenditure. These findings indicate that the methods are interchangeable not because the full-sample comparison is diluted by households on which they agree, but because the households over which they disagree experience similar welfare gains regardless of which method selects them.

Table 3: Impacts of Targeting on Downstream Outcomes among Marginal Households (ITT)

	FCS	rCSI	Food Exp. (USD)	Total Exp. (USD)	LCSI Stress (pp)	LCSI Crisis (pp)	LCSI Emer- gency (pp)	WHO- 5 Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Reference method: PMT-Exp</i>								
PMT-FCS	0.06 (0.62)	-0.69 (0.47)	7.18** (3.00)	17.03** (7.99)	1.12 (0.84)	3.78** (1.87)	-1.30 (1.05)	0.46 (0.72)
PMT-CARI	1.04* (0.62)	0.26 (0.48)	2.11 (3.04)	5.56 (8.02)	-0.08 (0.88)	1.52 (1.87)	0.00 (1.09)	0.49 (0.72)
Expert	-0.11 (0.62)	-0.22 (0.47)	4.54 (2.99)	14.75* (8.06)	1.05 (0.83)	-0.83 (1.87)	-1.84* (1.03)	0.18 (0.71)
Mean PMT-Exp.	54.60	16.82	135.61	391.22	94.33	56.50	8.97	22.63
Observations	5,566	5,566	5,566	5,566	5,566	5,566	5,566	5,517
p-val(PMT-FCS=PMT-CARI)	0.115	0.045	0.092	0.147	0.150	0.225	0.212	0.969
p-val(PMT-FCS=Expert)	0.783	0.316	0.372	0.774	0.924	0.013	0.586	0.694
p-val(PMT-CARI=Expert)	0.067	0.307	0.418	0.249	0.177	0.209	0.073	0.666

Notes: Sample restricted to *marginal* households—those selected by some but not all four targeting methods, excluding the always-eligible (selected by all four) and never-eligible (selected by none). PMT-Expenditure targeting serves as the reference group. All regressions include randomization-strata (eligibility-cell) fixed effects. Each column is its own dependent variable; the regressor of interest is the ITT treatment indicator (Round 1). Food and total expenditure are winsorized at the 95th percentile; LCSI coefficients and means are in percentage points; WHO-5 ranges 0–100. The p -values at the bottom test the null of equal treatment effects between targeting arms. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.4 Protective Role of Assistance During Conflict

The escalation of conflict between the two follow-up rounds allows us to examine whether continued assistance, and the targeting method that allocated it, helped households withstand a major shock. Around 56% of households lived in a locality that experienced at least one conflict event between survey rounds. We estimate how outcomes vary with conflict intensity and whether these effects differ by continued assistance and targeting arm.

Conflict had large and consistent effects on exposed households. A one-standard-deviation increase in conflict intensity raised the probability of displacement by 22.5 percentage points, shelter damage by 17.8 percentage points, and trauma (PTSD) symptoms by 1.17 points, all significant at the 1% level (Appendix Table A20). It also reduced the durable asset index by 0.14 standard deviations. Food-based coping (rCSI) and mental well-being (WHO-5) showed no

detectable average effects.

The protective role of continued assistance

Table 4 splits the conflict effect by whether a household had been selected to continue receiving assistance. Among the discontinued, the shock hit with full force—displacement, shelter damage, trauma, and asset losses all track the average effect. Continued assistance did not shield households from *physical* harm: the displacement and shelter-damage effects are essentially unchanged for continuers (interactions of -0.002 and -0.009 , both insignificant). It did, however, buffer *psychological* harm—the rise in trauma is 0.37 points smaller for households that kept their assistance ($p < 0.05$), about a quarter of the discontinued-group effect. Displacement is a function of where a household lives more than of its resources, whereas the income security that assistance provides appears to have cushioned the shock’s mental-health toll.

Table 4: Impact of conflict by continued vs discontinued assistance

Outcome	Discontinued	Continued	Difference
<i>Physical harm</i>			
Displaced	0.226*** (0.022)	0.224*** (0.022)	-0.002 (0.010)
Shelter damaged	0.183*** (0.021)	0.173*** (0.022)	-0.009 (0.010)
<i>Psychological</i>			
Trauma (PTSD)	1.358*** (0.294)	0.985*** (0.324)	-0.373** (0.186)
Mental well-being (WHO-5)	-0.393 (0.867)	0.510 (0.743)	0.904 (0.719)
<i>Asset depletion</i>			
Asset index	-0.154** (0.063)	-0.135 (0.095)	0.020 (0.067)
Sold productive assets	-0.011 (0.018)	0.023 (0.027)	0.034* (0.020)
<i>Coping</i>			
rCSI	-0.317 (0.581)	-0.276 (0.605)	0.040 (0.449)
LCSI stress (pp)	-1.072 (1.617)	-0.726 (1.623)	0.346 (1.405)
LCSI crisis (pp)	0.275 (1.858)	3.347* (1.938)	3.072* (1.642)
LCSI emergency (pp)	0.898 (0.716)	1.520 (0.993)	0.622 (0.916)

Notes: Estimates of the effect of conflict intensity—a one standard deviation increase in ACLED conflict intensity between Round 1 (mid-2024) and Round 2 (early 2025). The *Discontinued* column is the conflict effect among households whose assistance was discontinued; the *Continued* column is the effect among households selected to continue receiving assistance; and the *Difference* column is the gap between them (the conflict-intensity $\times$ continuation interaction). Outcomes observed only after the escalation (displacement, shelter damage, trauma, productive-asset sales) use a Round 2 cross-section with eligibility-cell fixed effects; repeated outcomes (WHO-5, asset index, rCSI, LCSI severity) use a two-period difference-in-differences with household fixed effects. LCSI severity indicators are in percentage points. Robust standard errors clustered at the locality level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Conflict intensity and outcomes by targeting arm

Table 5 examines whether the effects of conflict differ across targeting methods by interacting conflict intensity with the targeting arm, with PMT-Expenditure as the reference arm (column 1). Among PMT-Expenditure households, a one-standard-deviation increase in conflict intensity raised displacement by 23.5 percentage points, shelter damage by 18.6 percentage points, and trauma (PTSD) by 1.09 points (all $p < 0.01$), and lowered the durable-asset index by 0.16

standard deviations ($p < 0.05$); mental well-being and productive-asset sales showed no significant change. The arm interactions (columns 2–4) are generally small and statistically insignificant across all outcomes, indicating that the targeting method did not systematically affect households’ ability to withstand the shock. A more consistent pattern emerges for PMT-FCS. Relative to PMT-Expenditure, its households experienced smaller conflict-driven losses on several margins: displacement 2.5 percentage points lower ($p < 0.10$), productive-asset sales 4.6 percentage points lower ($p < 0.05$), and a 3.2 percentage-point smaller rise in stress-level coping ($p < 0.10$). The reference arm, PMT-Expenditure, exhibits the largest adverse responses across these outcomes, with the biggest increases in displacement, trauma, and emergency-level coping and the largest fall in the asset index, while on emergency coping PMT-CARI and the Expert rule also fare better than PMT-Expenditure (-2.2 points each, $p < 0.10$). These differences are individually marginal and spread across related outcomes, so we read them as suggestive rather than decisive. Still, they point in a consistent direction, with households targeted by PMT-FCS somewhat better insulated from the shock than those selected on expenditure. The protection is more pronounced among households that kept their assistance: restricting the PMT-FCS interaction to continuing beneficiaries, the reduction in displacement rises to 7.5 percentage points ($p < 0.01$) and shelter damage falls by 5.0 points ($p < 0.05$), as reported in Appendix Table [A21](#).

Table 5: Impact of conflict by targeting arm (relative to PMT-Exp)

Outcome	PMT-Exp (baseline)	PMT-FCS	PMT-CARI	Expert
<i>Physical harm</i>				
Displaced	0.235*** (0.025)	-0.025* (0.015)	-0.009 (0.016)	-0.004 (0.015)
Shelter damaged	0.186*** (0.022)	-0.008 (0.015)	-0.009 (0.016)	-0.016 (0.015)
<i>Psychological</i>				
Trauma (PTSD)	1.089*** (0.324)	0.255 (0.216)	-0.201 (0.264)	0.238 (0.237)
Mental well-being (WHO-5)	0.611 (0.744)	-0.102 (0.805)	-0.604 (0.879)	-1.282 (0.822)
<i>Asset depletion</i>				
Asset index	-0.157** (0.074)	-0.026 (0.060)	0.039 (0.068)	0.037 (0.069)
Sold productive assets	0.017 (0.022)	-0.046** (0.021)	0.002 (0.025)	-0.005 (0.022)
<i>Coping</i>				
rCSI	-0.144 (0.569)	-0.473 (0.601)	-0.070 (0.520)	-0.056 (0.468)
LCSI stress (pp)	0.597 (1.948)	-3.219* (1.827)	-3.105 (1.981)	0.359 (2.407)
LCSI crisis (pp)	2.347 (2.086)	-2.159 (2.322)	-0.157 (2.817)	0.918 (2.886)
LCSI emergency (pp)	2.825*** (1.029)	-1.893 (1.249)	-2.203* (1.326)	-2.162* (1.104)

Notes: Column 1 reports the baseline conflict impact—the effect of conflict intensity in the reference arm (PMT-Exp). Columns 2–4 report each arm’s additional effect relative to PMT-Exp (the conflict-intensity $\times$ arm interaction), so an arm’s total conflict effect is column 1 plus its own column. Conflict intensity is a one standard deviation increase in ACLED intensity between Round 1 (mid-2024) and Round 2 (early 2025). Repeated outcomes (WHO-5, rCSI, LCSI severity, asset index) use a two-period difference-in-differences with household fixed effects; outcomes observed only after the escalation (displacement, shelter damage, trauma, productive-asset sales) use a Round 2 cross-section with eligibility-cell fixed effects. LCSI severity indicators are in percentage points. Robust standard errors clustered at the locality level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Taken together, continued assistance offered real but bounded protection. It eased the conflict’s psychological toll without preventing displacement or physical damage. Among the deprivation-based methods, households targeted by PMT-FCS were somewhat better shielded, with smaller conflict-driven increases in displacement and productive-asset sales than under PMT-Expenditure. Whether an *impact-based* rule—one that prioritizes the households predicted to lose the most when assistance stops—would have protected welfare more effectively is the question we turn to next.

5.5 Targeting Vulnerability versus Impact

We next compare the deprivation-based methods with an impact-based rule that ranks households by their predicted welfare loss from assistance discontinuation, considering separate rankings based on total and food expenditure. The two approaches select materially different households. They overlap on only 65% of selected households when based on total expenditure and 61% when based on food expenditure, while the corresponding household rankings are negatively correlated ($\rho = -0.31$ and -0.42). Compared with the PMTs, the impact-based rule selects households that are closer to the caseload average in assets and demographic characteristics (Appendix Table A22), indicating that current deprivation and expected losses from assistance withdrawal capture distinct dimensions of need.

To compare the targeting rules on a common scale, Table A23 reports the share of first-best social welfare¹⁸ captured by each method, holding the distributional parameter fixed at $\rho = 3$. The impact-based rules outperform all deprivation-based approaches by a wide margin. Impact-Total captures 0.91 of the first-best welfare gain for total expenditure and Impact-Food captures 0.85 for food expenditure, compared with 0.64–0.69 for the deprivation-based PMTs. The Expert rule performs worst among the targeting methods (0.62–0.64), while random allocation provides a lower bound (0.52–0.54). Each impact-based rule performs best for the expenditure measure on which it is trained while also performing well on the alternative measure. Neither, however, achieves the first-best benchmark, which places greater weight on welfare gains among poorer households than the predicted-loss ranking alone.

Appendix Table A24 helps explain why the deprivation-based methods deliver similar welfare despite selecting different households. None of the three PMTs meaningfully ranks households by their welfare gain from continued assistance. The rank correlation between selection and household-level gains ranges from only -0.04 to 0.03 and is similar across methods. As a result, each captures approximately the average welfare gain in the caseload, making the three PMTs close substitutes in terms of welfare. By contrast, the impact-based rule is strongly correlated with household-level gains ($\rho = 0.70$) and captures roughly four-fifths of the first-best welfare gain. The deprivation-based methods achieve moderate welfare not by identifying households with the largest gains from assistance, but by consistently selecting poor households, who receive greater weight under the social welfare function. Because all three PMTs target similarly poor households, they generate similar welfare despite selecting different beneficiaries.

The impact arms' advantage is specific to the expenditure outcomes they are built to protect, and does not extend to the other dimensions of welfare. Repeating the welfare-capture exercise on the conflict and well-being outcomes of Section 5.4—food security, reduced-coping, trauma, mental well-being, and assets—the impact arms no longer lead: on reduced coping the deprivation-based methods capture more of the attainable gain than the impact arms do, and trauma and mental well-being are close to untargetable by any rule, with random allocation no worse than the rest (Appendix Table A25). Ranking on expenditure loss buys welfare where

¹⁸The first-best is the welfare-maximizing feasible allocation—the assignment that, with perfect foresight of each household's welfare gain, directs the same 43% of assistance to the households where it raises social welfare most under the CRRA objective. The welfare-capture ratio G_m is the share of this maximum attainable gain, relative to providing no assistance, that method m achieves, so random allocation lies well above zero and the first-best equals one.

expenditure is the outcome; it does not generalize to harms, such as displacement or psychological distress, that are weakly related to a household’s expenditure response. The practical lesson is not that one rule dominates, but that a data-driven method can be pointed at whichever outcome a program most wants to protect—at no additional data-collection cost—whereas a fixed deprivation or Expert rule cannot.

5.6 Cost-Effectiveness

The welfare comparison so far abstracts from implementation costs. With the transfer budget fixed, the main difference across targeting methods is the cost of collecting the information they require, so the relevant comparison is welfare net of data-collection costs (Aiken et al., 2025).

The methods differ substantially in their data requirements. The PMTs require a household survey collecting a broad set of indicators, whereas the Expert rule relies on a small number of administrative characteristics and can be implemented through an existing population registry where one is available. The impact-based rule uses the same survey data as the PMTs, requiring only an additional analytical step to estimate treatment effects. As a result, switching from deprivation-based to impact-based targeting entails essentially no additional data-collection cost.

Appendix Table A26 incorporates each method’s data-collection costs into the welfare comparison. Re-surveying households for the PMTs and the impact-based rule costs about 4.4% of the transfer budget per cycle, while maintaining an expert registry costs 1.5–3.0% once periodic updates are spread over time. Because these costs are small relative to transfers, the welfare ranking changes little. The impact-based rule remains the most cost-effective ($G_m^{adj} = 0.91$), and the ordering among the deprivation-based methods is unchanged, with the PMTs continuing to outperform the Expert rule despite their higher data costs. The Expert approach therefore becomes cost-advantageous only when a population registry already exists. Otherwise, the additional survey costs are small relative to the welfare gains achieved by the data-driven methods.

6 Conclusions

This paper provides experimental evidence on the performance of alternative targeting methods during the scale-down of a large humanitarian assistance program. Exploiting the randomized implementation of four targeting approaches in Lebanon, we compare whom each method selects and how the populations they prioritize fare following the discontinuation of assistance and the subsequent conflict escalation. We also compare targeting based on current deprivation with targeting based on predicted program impact.

The four targeting methods produced systematically different selection patterns. The three PMT models—based on predicted expenditure, food consumption, or multidimensional vulnerability—overlapped only partially in identifying households most in need. The Expert approach, relying on expert-defined indicators of vulnerability, selected a markedly different profile, including a higher share of female-headed households, unemployed individuals, and persons with disabilities. These results show that each method captures distinct aspects of vulnerability, resulting in meaningful differences in household selection.

None of the methods consistently generated better outcomes across all indicators. PMT-CARI showed a modest advantage in food security, but overall household outcomes did not differ significantly by targeting method. The clearest subgroup differences arose for elderly-headed households and the poorest households (the bottom asset quartile), which fared somewhat better under PMT-FCS and the Expert rule. More broadly, no method clearly favored or disadvantaged specific population subgroups in average welfare.

The randomized design also lets us look past selection to the welfare consequences of the targeting rule. Two findings stand out. When conflict escalated between the survey rounds, continued assistance buffered the psychological toll of the shock—trauma rose markedly less among households that kept their support—but did not prevent displacement or physical damage, and which deprivation-based method had delivered the assistance made no discernible difference. And in a social-welfare simulation, a rule that ranks households by their predicted loss from discontinuation captures about nine-tenths of the first-best welfare gain on expenditure, against six- to seven-tenths for the proxy-means tests and least of all for the Expert rule.

These findings illustrate that the optimal targeting rule depends on the planner’s objective. The households who are most deprived today are not necessarily those who lose the most when assistance is withdrawn, and the deprivation- and impact-based rules overlap on only about two-thirds of the households they select. As a result, deprivation-based approaches may overlook vulnerable groups, such as some elderly-headed households, requiring complementary safeguards when assistance is scaled back.

For policy, the two families of methods are complements more than rivals. A data-driven method can model several outcomes at once—current deprivation, food insecurity, or expected loss from withdrawal—and be re-pointed at whichever a program adopts, all at no additional data-collection cost. An expert or registry-based rule can be cheaper to administer where a population registry already exists, but in our setting it ranks the deprived less accurately and delivers lower welfare. The right choice turns on what data an agency already holds and what it is trying to maximize, not on a presumption that one family dominates everywhere.

These conclusions carry caveats. The targeting outcomes were not recorded in the census used to assign households, so accuracy is validated using the training database rather than the assignment census; the impact-based arm is evaluated by simulation rather than implemented in the field; and the conflict and the emergency cash scale-up that followed were both unanticipated, with the scale-up reaching most discontinued households and so muting any differences in realized welfare across arms.

The findings highlight both the importance of targeting choices in shaping who is reached during program contraction and the significant welfare costs of discontinuing assistance. The results demonstrate that different targeting methods capture distinct dimensions of vulnerability, but no approach yields consistently superior outcomes. At the same time, the termination of aid generates sizable welfare losses, particularly for the poorest and most vulnerable households, with limited evidence of effective self-preparation for exit. These insights underscore the need for careful consideration of both targeting and exit strategies in humanitarian programming.

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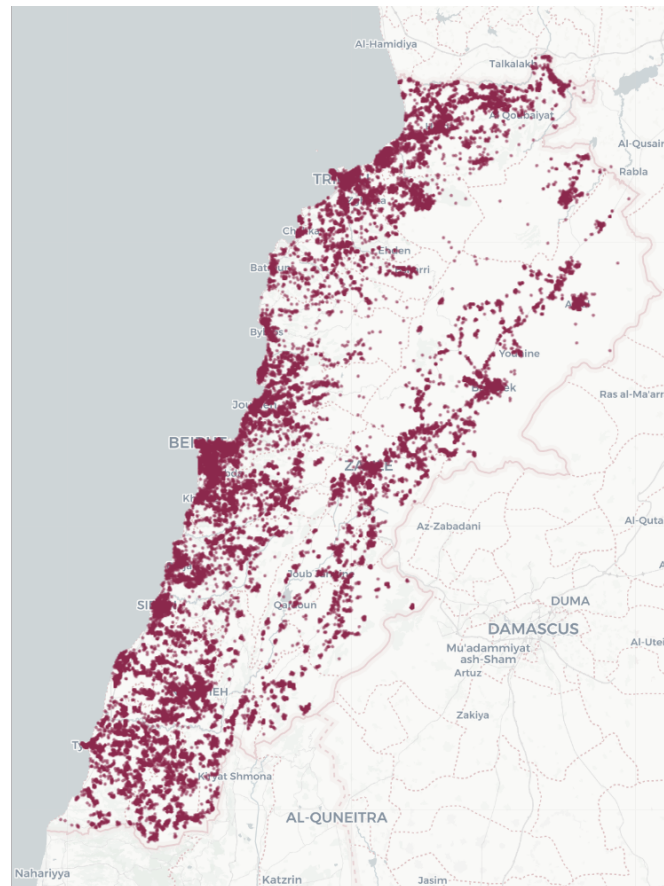
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Appendix

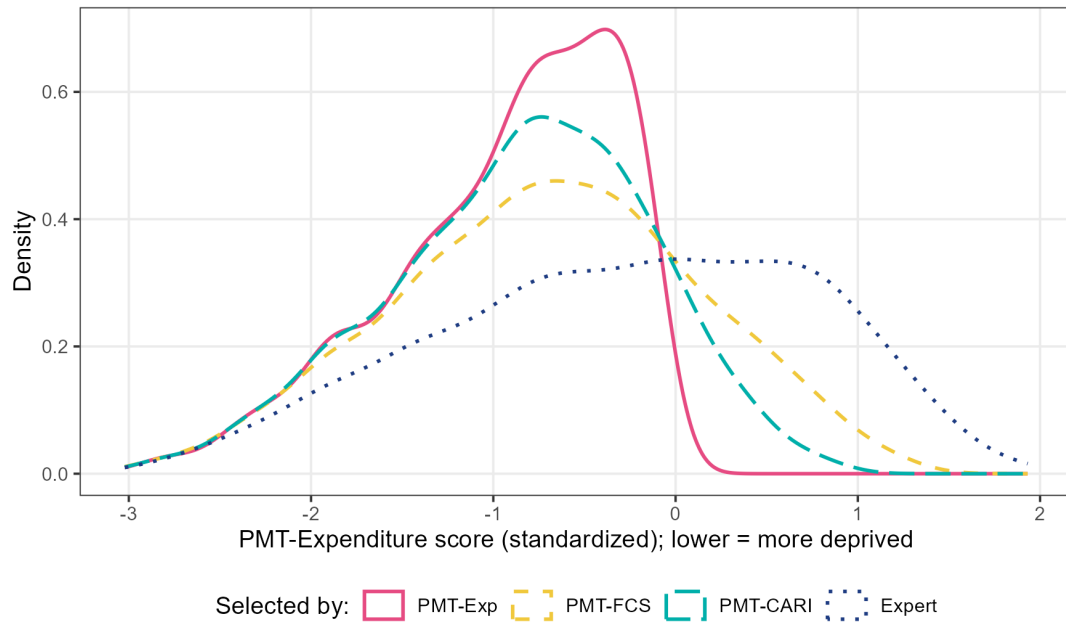
A Additional figures

Figure A1: Distribution of 2023 food assistance beneficiaries across Lebanon



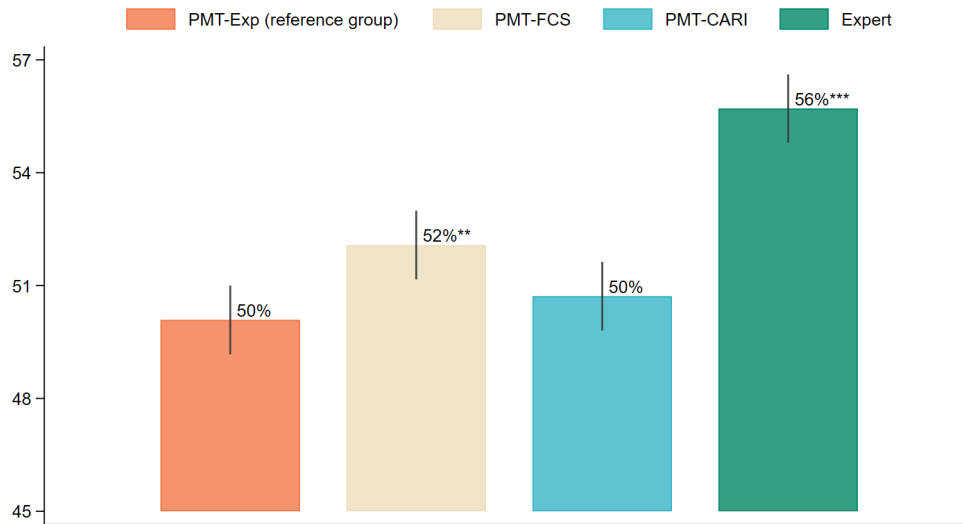
Note: Own visualization. The map displays the geographic locations of households receiving assistance from WFP in 2023.

Figure A2: PMT-Expenditure score by selection status across targeting methods



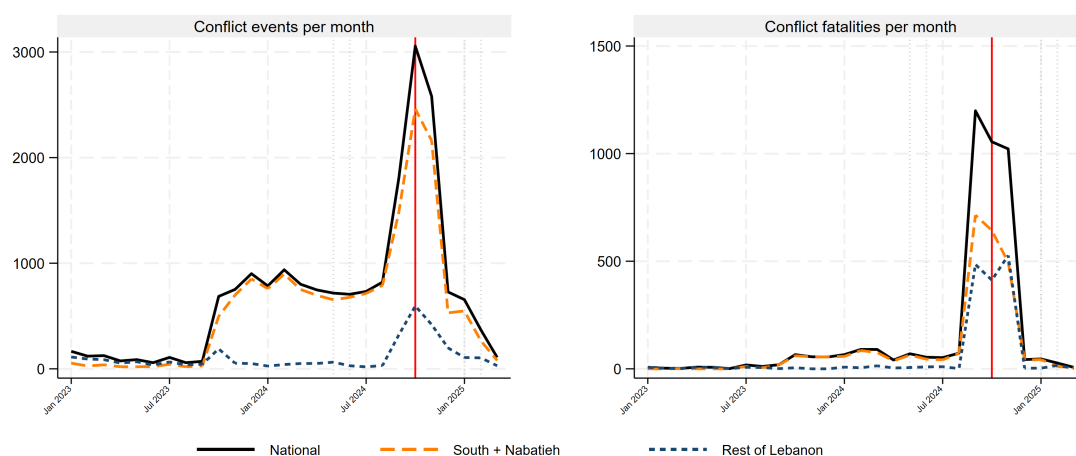
Note: Kernel densities of the standardized PMT-Expenditure score among the 43% of households each method selects for continued assistance. Lower scores denote households predicted to be more deprived. PMT-Expenditure selects the lowest-scoring households by construction, whereas the Expert rule selects across the distribution, including households that are not expenditure-poor.

Figure A3: Appeal rates among discontinued households, by targeting method



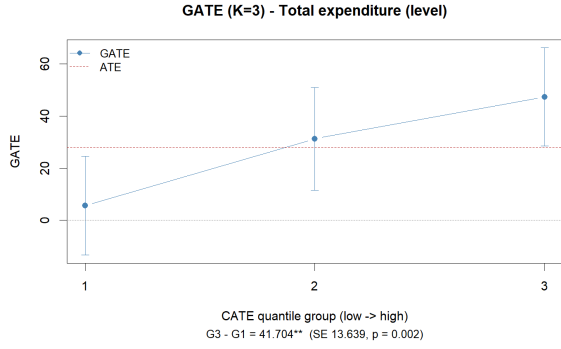
Notes: Share of discontinued households that lodged an appeal against the discontinuation decision, by targeting arm, from the administrative appeals records over the full beneficiary census. PMT-Expenditure is the reference arm, and stars denote significance relative to it. ** $p < 0.05$, *** $p < 0.01$.

Figure A4: Conflict escalation between survey rounds: ACLED events and fatalities per month

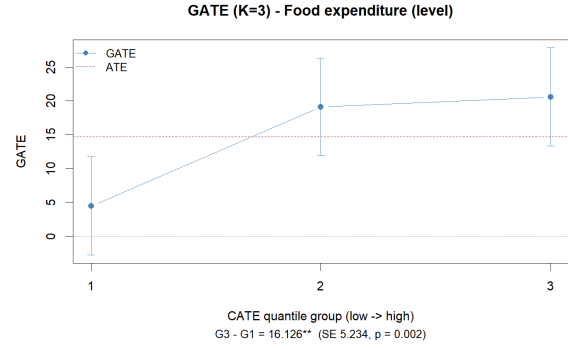


Notes: Monthly ACLED conflict events (left) and fatalities (right) in Lebanon, nationally and split between the south (South + Nabatieh) and the rest of the country. The red line marks the October 2024 escalation, which fell between the two survey rounds; the dotted gray lines mark the survey windows. The escalation was concentrated in the south. This is the variation the difference-in-differences design of Section 4 exploits.

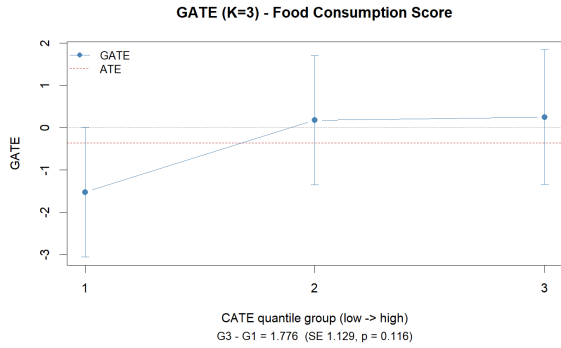
Figure A5: Group average treatment effects (GATES) of discontinuation by tertile of the predicted effect



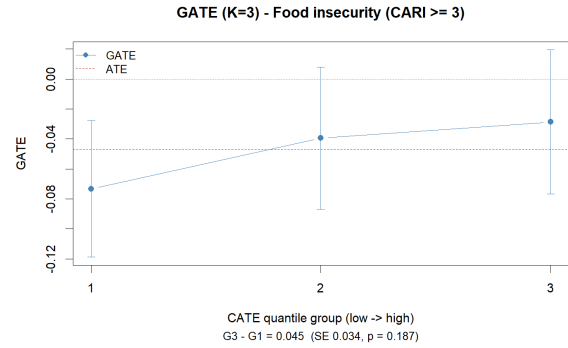
(a) Total expenditure



(b) Food expenditure



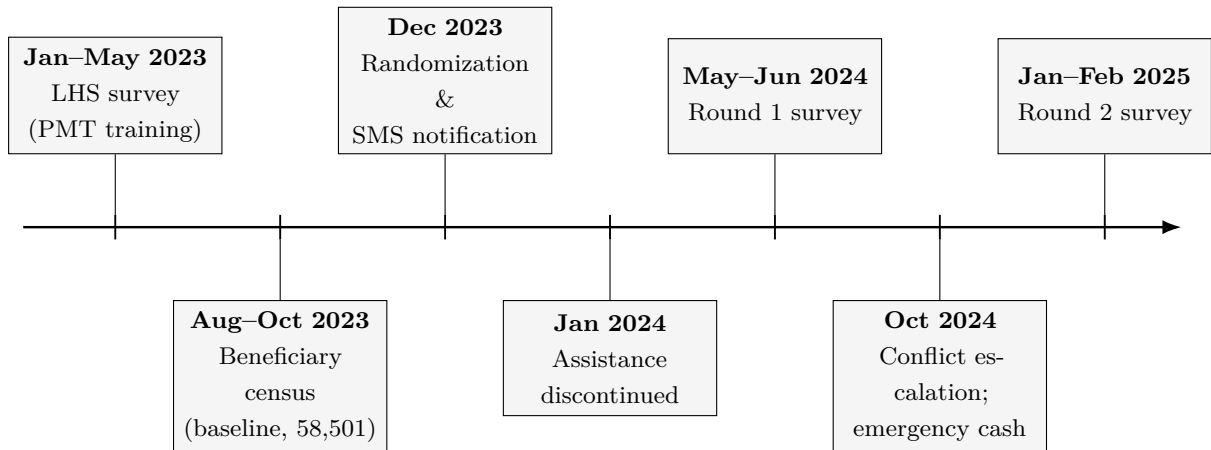
(c) Food Consumption Score



(d) Food insecurity (CARI ≥ 3)

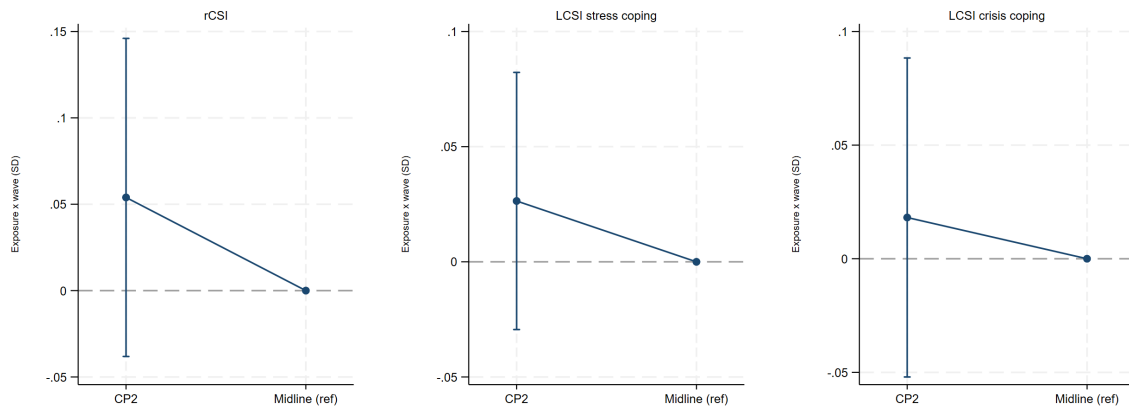
Note: Each panel plots the group average treatment effect of discontinuation within tertiles of the household-level predicted effect (CATE), estimated by causal forest (Wager and Athey, 2018) within the generic machine-learning framework of Chernozhukov et al. (2018) ($N = 6,147$; 2,049 households per tertile). Whiskers are 95% confidence intervals; the dashed line marks the average treatment effect. The top and bottom tertiles differ significantly for the expenditure outcomes but not for the food-consumption outcomes.

Figure A6: Study timeline



Notes: The Lebanon Household Survey (LHS) trains the PMT models; the WFP beneficiary census provides the baseline universe of 58,501 households. Households were assigned to targeting arms and notified of their 2024 status by SMS, and the 57% not retained were discontinued in January 2024. Round 1 took place before, and Round 2 after, the escalation of conflict in October 2024, when WFP began emergency cash transfers in affected areas.

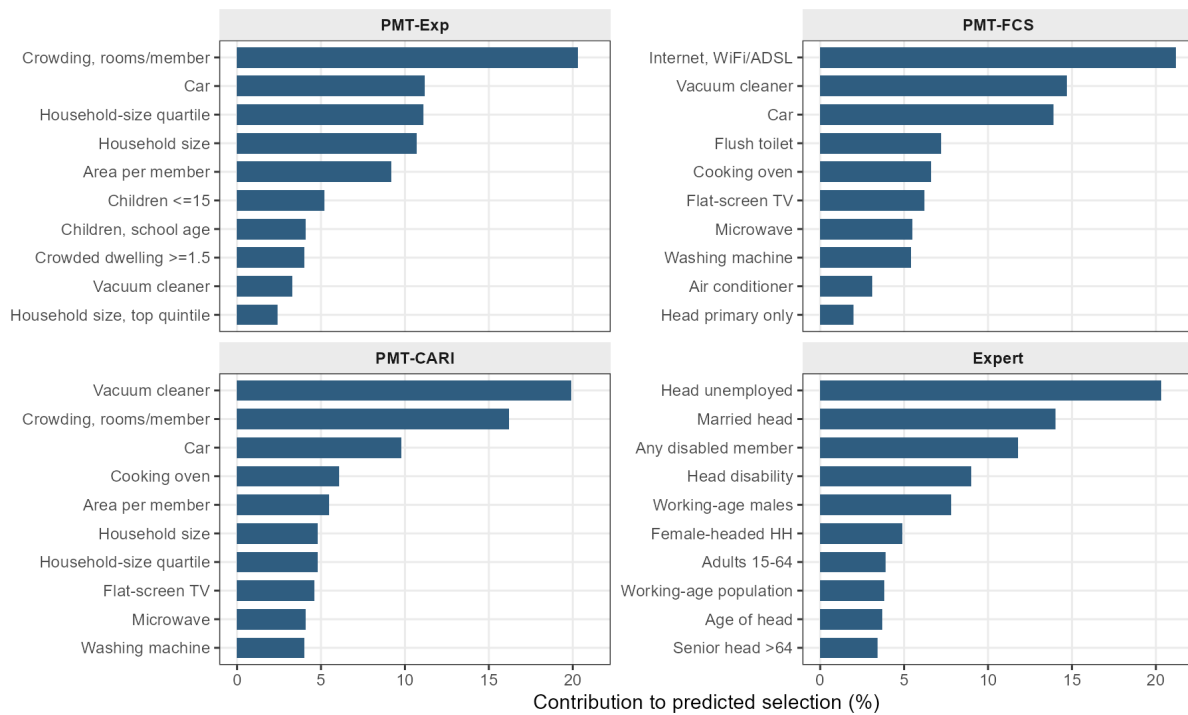
Figure A7: Pre-conflict parallel-trends check: exposure $\times$ wave by outcome



Pre-conflict exposure $\times$ wave coefficient (per SD of ACLED intensity), relative to midline. CP2 = late 2023. CP2 indistinguishable from zero = parallel pre-trends.

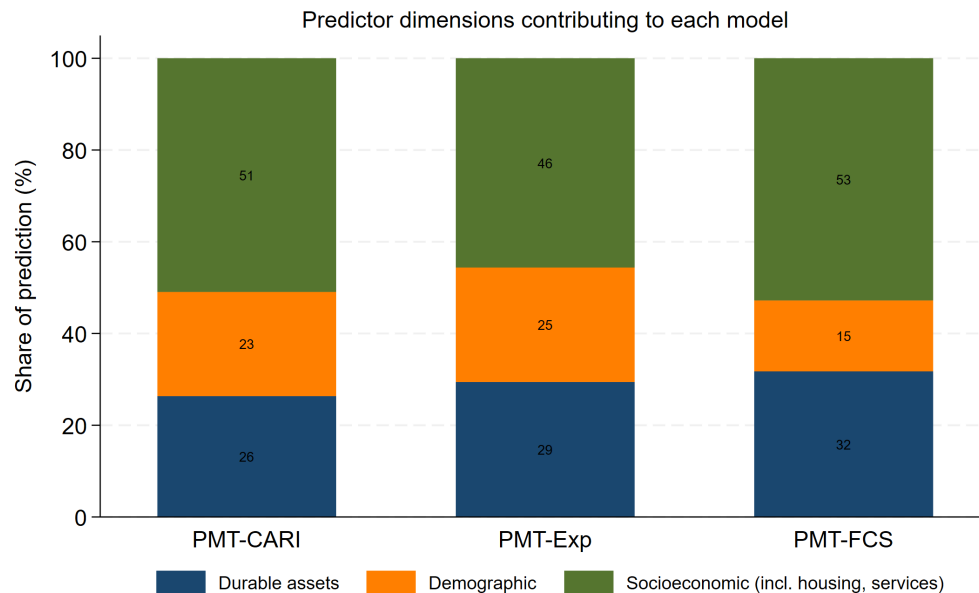
Notes: Each panel plots the coefficient on conflict exposure (per standard deviation of ACLED intensity) interacted with survey wave, relative to Round 1 (reference, set to zero), for the two pre-conflict rounds—an earlier version of the census (late 2023) and Round 1 (mid-2024). Vertical bars are 95% confidence intervals (locality-clustered). An earlier-round coefficient indistinguishable from zero indicates that more- and less-exposed households were on a common trajectory before the escalation. The Round 2 (post-conflict) effect is the treatment effect of interest and is reported in the impact tables, not here.

Figure A8: Predictor contributions to selection, by targeting method



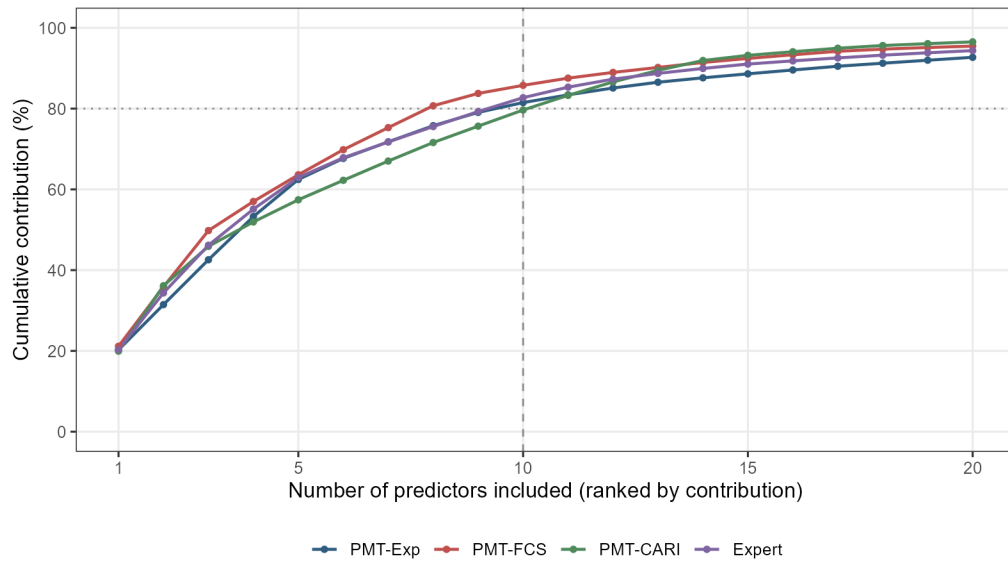
Notes: Each panel shows the ten predictors contributing most to a method's predicted selection. Contribution is the absolute penalized LASSO coefficient times the predictor's standard deviation, normalized to sum to 100% within each model (Appendix Table A12).

Figure A9: Predictor dimensions contributing to each PMT



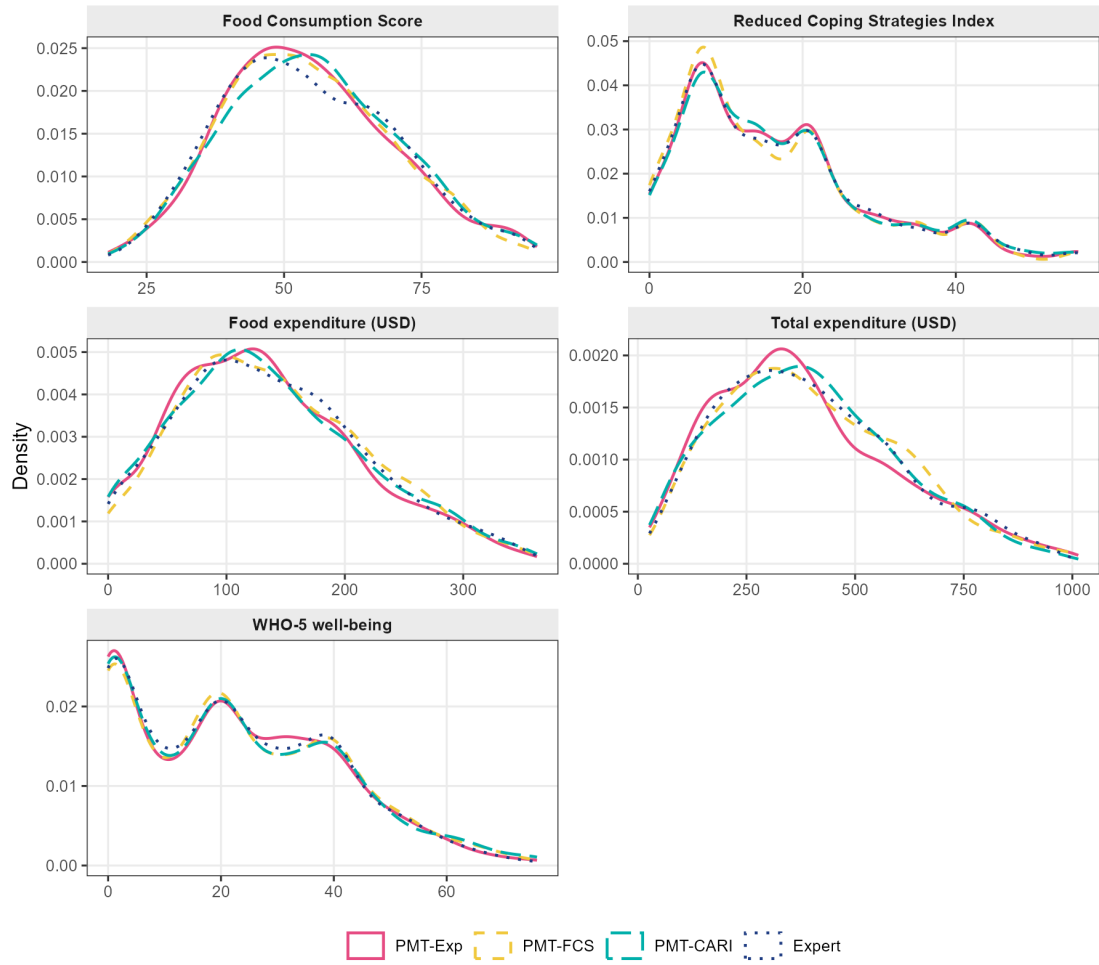
Notes: Share of each LASSO model's prediction attributable to each covariate group, where a covariate's contribution is the absolute penalized coefficient times the covariate's standard deviation, normalized to sum to 100% within each model. Covariates are grouped into durable assets, demographic characteristics, and socioeconomic conditions (the latter including housing and access to services).

Figure A10: Cumulative predictor contribution by number of predictors retained



Notes: For each method, the cumulative contribution of predictors ranked by importance (contribution defined as in Appendix Figure A8). The dashed line marks the tenth predictor and the dotted line marks 80%; the top ten predictors account for 80–86% of the total contribution in every model.

Figure A11: Distribution of downstream outcomes by targeting method



Note: Kernel densities of the downstream outcomes—Food Consumption Score, reduced Coping Strategies Index (rCSI), food expenditure, total expenditure, and WHO-5 well-being—in Round 1, disaggregated by targeting method (whole study population, beneficiaries and non-beneficiaries). Each outcome is trimmed at its 1st and 99th percentiles for display. The distributions are nearly identical across the four targeting arms.

B Additional tables

Table A1: Composition and caloric contribution of the food basket

Commodity	Food group	Quantity	kcal contribution ^{1/}	Contribution to basket
Egyptian Rice	Cereals	3 KG	10,800	47%
Brown Bulgur	Cereals	3 KG	10,260	
Pasta	Cereals	4 KG	14,840	
Lentils	Pulses	4 KG	13,720	27%
White Beans	Pulses	1 KG	3,330	
Chickpeas	Pulses	1 KG	3,640	
Sunflower Oil	Oils & Fat	1.84 ltr.	15,000	20%
Salt	Condiments	500 g	0	0%
Tomato Paste	Condiments	400 g	328	
Sugar	Sugar	1 KG	3,870	5%

Note: The contribution represents kcal equivalencies of the defined quantity of a basket for two people.

Table A2: Correlation in selection across targeting methods

	PMT-Exp	PMT-FCS	PMT-CARI	Expert
PMT-Exp	1.00			
PMT-FCS	0.41	1.00		
PMT-CARI	0.65	0.59	1.00	
Expert	0.02	0.18	0.12	1.00

Notes: Pairwise correlations between binary selection indicators (whether a household falls in the selected 43%) across the four targeting methods, computed over the whole beneficiary population ($N = 58,501$). The average correlation is 0.33 across all six pairs and 0.55 among the three PMT-based methods.

Table A3: Sample distribution across governorates

Governorate	Sample		Population	
	N	%	N	%
Akkar	729	11.5	7,140	12.2
Baalbek-El Hermel	596	9.4	4,241	7.2
Beirut	363	5.7	2,941	5.0
Bekaa	492	7.7	4,156	7.1
El Nabatieh	493	7.8	6,208	10.6
Mount Lebanon	1,807	28.5	16,928	28.9
North	1,076	16.9	9,627	16.5
South	794	12.5	7,260	12.4
Total	6,350	100.0	58,501	100.0

Notes: The “Sample” columns describe the analysis sample ($N = 6,350$); the “Population” columns describe the full WFP beneficiary census of 58,501 households from which the sample was drawn and over which targeting eligibility was defined. For each governorate the table reports the number of households (N) and its share of the respective total (%). Governorate is measured at baseline, before conflict-induced displacement. Because the sample is drawn by eligibility cell rather than by location, its geographic shares track but do not exactly match those of the population.

Table A4: Out-of-sample modeling comparison: forward-stepwise OLS, Ridge, Elastic Net, LASSO, and GRF

Method	Consumption (log p.c.)			Food Consumption Score			Food insecurity (CARI > 2)		
	MSE	R^2	#cov	MSE	R^2	#cov	MSE	D^2	#cov
Stepwise OLS	0.312	0.508	51	639.8	0.157	27	0.134	0.165	31
Ridge	0.307	0.515	192	638.2	0.159	192	0.127	0.256	192
Elastic Net	0.307	0.516	118	636.9	0.161	93	0.126	0.260	97
LASSO	0.307	0.516	117	636.6	0.161	92	0.126	0.261	92
GRF	0.355	0.440	192	649.1	0.145	192	0.133	0.226	192

Notes: Models are fit on the 75% training split of the LHS and evaluated on the held-out 25% test split (untrimmed sample). For the continuous outcomes (consumption, FCS) we report mean squared error and out-of-sample R^2 ; for the binary food-insecurity outcome (CARI > 2) we report the Brier score (MSE) and the deviance ratio D^2 . OLS uses forward stepwise selection (a linear probability model for the binary outcome); Ridge, Elastic Net, and LASSO are regularized regressions (logit for the binary outcome); GRF is a generalized random forest. # cov. is the number of covariates selected (Ridge and GRF use all available covariates).

Table A5: Detailed description of the indices

Indicator	Components	Sub-components	Possible answers	Weight Score range
Food Consumption Score (FCS)	Weighted frequency of food groups consumed over past 7 days	Cereals, roots, and tubers	0 - 7 days	2
		Pulses	0 - 7 days	3
		Vegetables	0 - 7 days	1
		Fruits	0 - 7 days	1
		Meat and fish	0 - 7 days	4
		Milk	0 - 7 days	4
		Sugar	0 - 7 days	0.5
		Oil	0 - 7 days	0.5
Reduced Coping Strategy Index (rCSI)	Frequency of 5 coping strategies used in past 7 days	Rely on less preferred or less expensive foods	0 - 7 days	1
		Borrow food or rely on help from a friend or relative	0 - 7 days	2
		Limit portion size at mealtimes	0 - 7 days	1
		Restrict consumption by adults so children can eat	0 - 7 days	3
		Reduce number of meals eaten in a day	0 - 7 days	1
Livelihood Coping Strategy Index (LCSI)	Whether the household adopted coping strategies since January 2024, grouped by severity	<i>Stress</i> : reduce spending on fuel, water, or electricity	No; Yes; Not applicable	n.a.
		<i>Stress</i> : sell household assets (furniture, fridge, TV, phone, jewelry)	No; Yes; Not applicable	n.a.
		<i>Stress</i> : move children to a less expensive school	No; Yes; Not applicable	n.a.
		<i>Stress</i> : reduce non-essential health expenditure	No; Yes; Not applicable	n.a.
		<i>Crisis</i> : sell productive, income-generating assets (e.g., sewing machine, vehicle)	No; Yes; Not applicable	n.a.
		<i>Crisis</i> : withdraw children from school	No; Yes; Not applicable	n.a.
		<i>Crisis</i> : postpone or stop essential or chronic medical treatment	No; Yes; Not applicable	n.a.
		<i>Emergency</i> : sell an owned house or land	No; Yes; Not applicable	n.a.
		<i>Emergency</i> : send household members to beg	No; Yes; Not applicable	n.a.
		<i>Emergency</i> : adults engage in illegal or high-risk work	No; Yes; Not applicable	n.a.

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Table A5 continued from previous page

Indicator	Components	Sub-components	Possible answers	Weight	Score range
WHO-5 Mental Health Index	Sum of 5 items from WHO-5 well-being scale	I have felt cheerful and in good spirits	1 = Never to 5 = All of the time; -999 = Prefer not to disclose	4	0-100
		I have felt calm and relaxed	1 = Never to 5 = All of the time; -999 = Prefer not to disclose	4	
		I have felt active and vigorous	1 = Never to 5 = All of the time; -999 = Prefer not to disclose	4	
		I woke up feeling fresh and rested	1 = Never to 5 = All of the time; -999 = Prefer not to disclose	4	
		My daily life has been filled with things that interest me	1 = Never to 5 = All of the time; -999 = Prefer not to disclose	4	
Trauma / PTSD Index (ITQ, ICD-11)	Sum of 6 symptom items, bothered by each in the past month (Cloitre et al. 2018)	Upsetting dreams that replay part of the experience	0 = Not at all to 4 = Extremely; -999 = Prefer not to disclose	1	0-24
		Powerful images/memories as if the experience is happening again	0 = Not at all to 4 = Extremely; -999 = Prefer not to disclose	1	
		Avoiding internal reminders (thoughts, feelings, sensations)	0 = Not at all to 4 = Extremely; -999 = Prefer not to disclose	1	
		Avoiding external reminders (people, places, activities)	0 = Not at all to 4 = Extremely; -999 = Prefer not to disclose	1	
		Being “super-alert”, watchful, or on guard	0 = Not at all to 4 = Extremely; -999 = Prefer not to disclose	1	
		Feeling jumpy or easily startled	0 = Not at all to 4 = Extremely; -999 = Prefer not to disclose	1	

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Indicator	Components	Sub-components	Possible answers	Weight Score range	
Asset Index	Ownership of durable goods (first principal component, PCA)	Refrigerator; separate freezer; cooking burner with oven; cooking burner without oven; independent electric oven; microwave; automatic washing machine; dishwasher; dryer; vacuum cleaner; flat-screen TV; other TV; DVD/Blu-ray player; satellite dish; air conditioner; water heater; computer/laptop; smartphone; other mobile phone; tablet; WiFi internet; mobile internet; water purifier; water pump; medical appliance; motorcycle/scooter; ATV/jet-ski; bus/mini-bus; pickup truck; bicycle; boat; tuk-tuk; car	Owned (1) / not owned (0)	PCA load-ings	Standardized (z)

Notes: Construction of the well-being and food-security indices. Items marked “(reversed)” are recoded so that higher index values denote worse outcomes; “Possible answers” lists the raw response scale, with $-999/-99$ codes treated as missing. The FCS and rCSI use the standard WFP weights; the WHO-5 sum is rescaled to 0–100. “n.a.” indicates an unweighted (equal-weight) sum of the listed sub-components. The Trauma/PTSD index is the summed severity of the six ICD-11 symptom items of the International Trauma Questionnaire (Cloitre et al., 2018); the three functional-impairment items are used only for the caseness definition. The asset index is the first principal component of the durable-good ownership indicators, so its “weights” are PCA loadings and its score is standardized rather than a bounded sum.

Table A6: Conflict exposure by geography, targeting arm, and eligibility

	N	Exp. rate	Mean ev.	Mean fat.	%None	%Low	%Med	%High
Panel A: By governorate								
Akkar	838	0.26	0.8	0.0	74.5	21.1	4.4	0.0
Baalbek-El Hermel	690	0.89	22.0	17.4	11.0	19.3	33.5	36.2
Beirut	426	0.95	11.5	22.6	5.2	28.4	46.5	20.0
Bekaa	578	0.41	2.0	0.8	59.2	33.9	4.0	2.9
El Nabatieh	571	0.97	28.3	10.9	3.0	1.9	24.7	70.4
Mount Lebanon	2122	0.53	19.2	4.3	47.1	18.7	2.3	32.0
North	1245	0.22	1.0	0.2	77.7	9.3	13.0	0.0
South	929	0.80	22.2	10.9	20.0	7.4	27.1	45.4
South + Nabatieh	1500	0.86	24.5	10.9	13.5	5.3	26.2	54.9
Panel B: By targeting arm								
PMT-Exp	1855	0.56	14.2	6.2	43.9	14.7	14.8	26.6
PMT-FCS	1842	0.58	13.7	6.6	42.4	17.8	14.7	25.1
PMT-CARI	1839	0.57	13.4	6.4	42.9	16.6	16.3	24.3
Expert	1863	0.54	13.1	6.7	45.5	17.0	13.3	24.2
Panel C: By eligibility space								
e0	441	0.55	11.3	5.5	44.9	20.2	12.9	22.0
e1	497	0.61	14.4	5.7	38.6	18.1	13.5	29.8
e1e2	443	0.66	14.7	7.9	34.3	18.3	19.6	27.8
e1e2e3	463	0.47	7.2	3.5	53.1	16.8	17.1	13.0
e1e2e3e4	442	0.50	10.2	4.9	50.2	17.0	15.4	17.4
e1e2e4	434	0.67	16.3	10.1	33.4	17.5	20.0	29.0
e1e3	477	0.57	15.9	4.9	43.4	11.1	16.1	29.4
e1e3e4	453	0.59	19.1	7.5	41.3	15.2	9.9	33.6
e1e4	461	0.60	18.8	8.3	39.7	15.4	13.7	31.2
e2	492	0.59	10.6	7.4	41.1	19.3	16.9	22.8
e2e3	468	0.48	8.8	5.8	51.7	19.2	14.3	14.7
e2e3e4	467	0.55	14.6	7.7	45.0	15.6	15.6	23.8
e2e4	474	0.69	15.8	10.2	31.0	14.3	24.3	30.4
e3	451	0.45	11.5	3.7	55.0	14.4	6.9	23.7
e3e4	441	0.49	15.8	4.6	51.0	15.0	7.7	26.3
e4	495	0.54	12.9	6.0	45.9	16.4	11.9	25.9

Notes: Conflict exposure from ACLED (Oct 2024–Feb 2025) merged to households by cadaster P-code. The exposure rate is the share of households in a locality with any event; mean events/fatalities are locality values; %None–%High give the dose distribution. Dose bins are event tertiles among exposed localities (low 1–3, medium 4–16, high 17–123). † marks cells with N<30.

Table A7: Balance in randomization (study arm assignment on baseline covariates)

	Coef. of selection (robust SE)			
	PMT-Exp	PMT-FCS	PMT-CARI	Expert
Female headed household	0.001 (0.004)	0 (0.004)	0.002 (0.004)	-0.003 (0.004)
Age of the household head	-0.052 (0.134)	-0.085 (0.133)	0.163 (0.133)	-0.027 (0.133)
Household head divorced/separated or widowed	-0.003 (0.004)	0.003 (0.004)	0.004 (0.004)	-0.004 (0.004)
HHH is married	0 (0.004)	-0.001 (0.004)	0.001 (0.004)	0.001 (0.004)
Household head never attended school	0.004 (0.004)	0 (0.004)	-0.006 (0.004)	0.002 (0.004)
Household head with tertiary education	0.004 (0.002)	0 (0.002)	-0.001 (0.002)	-0.003 (0.002)
Number of HH members	0.009 (0.017)	-0.003 (0.017)	0.01 (0.017)	-0.016 (0.017)
Dependency ratio (nonproductive/hhmembers)	-0.001 (0.003)	-0.001 (0.003)	0.002 (0.003)	-0.001 (0.003)
At least one disabled HH member	0.005 (0.004)	-0.008* (0.004)	0.001 (0.004)	0.002 (0.004)
At least one child attends private school	-0.006* (0.004)	0.001 (0.004)	0.005 (0.004)	0 (0.004)
Skilled regular job private	0 (0.002)	0.002 (0.002)	0.002 (0.002)	-0.004*** (0.001)
Business owner informal	-0.001 (0.003)	-0.004 (0.003)	0.001 (0.003)	0.003 (0.003)
Unemployed	-0.005 (0.004)	0.005 (0.004)	-0.003 (0.004)	0.003 (0.004)
No syrians in HH	-0.002 (0.003)	-0.001 (0.003)	0.006 (0.004)	-0.003 (0.003)
HH does not own house	-0.005 (0.005)	0.003 (0.005)	0.001 (0.005)	0.002 (0.005)
dwelling area per HH member	-0.168 (0.269)	0.163 (0.269)	-0.044 (0.267)	0.049 (0.268)
Flush toilet	-0.001 (0.003)	0.004 (0.003)	0.002 (0.003)	-0.005 (0.003)
Piped water access	0.002 (0.004)	0 (0.004)	0.002 (0.004)	-0.004 (0.004)
Use gas heating	-0.001 (0.003)	0.002 (0.003)	0.003 (0.003)	-0.004 (0.003)
Use gas for cooking	0.002 (0.004)	-0.002 (0.004)	-0.002 (0.004)	0.002 (0.004)
Use electricity from own solar panel	-0.004 (0.003)	0.006** (0.003)	-0.001 (0.003)	-0.001 (0.003)
WiFi Internet (e.g. Cable / ADSL)	0.004 (0.005)	0.004 (0.005)	0.001 (0.005)	-0.008* (0.005)
Computer / Laptop	0 (0.003)	0 (0.003)	0.001 (0.003)	-0.001 (0.003)
Mobile Internet	0.005 (0.005)	-0.003 (0.005)	-0.003 (0.005)	0.001 (0.005)
Flat screen TV	-0.007 (0.005)	0.002 (0.005)	0.001 (0.005)	0.004 (0.005)
Vacuum cleaner	0 (0.004)	0.004 (0.004)	-0.005 (0.004)	0.001 (0.004)
Microwave	0.002 (0.004)	0.005 (0.004)	-0.001 (0.004)	-0.005 (0.003)
Air conditioner	0 (0.004)	0.006 (0.004)	0.003 (0.004)	-0.008** (0.004)
Scooter / Moped / Motorcycle	0.001 (0.003)	0.006** (0.003)	-0.003 (0.003)	-0.004 (0.003)
Car	0 (0.005)	-0.003 (0.005)	0.001 (0.005)	0.003 (0.005)

Notes: Each cell reports the coefficient (robust SE in parentheses) from a separate OLS regression of an indicator for assignment to the given targeting arm on the listed baseline covariate, estimated over the full beneficiary census ($N = 58,501$). The rows shown are a representative subset of the baseline covariates. Across the full set of 153 baseline variables, 7.8% of the p-values fall below 0.10, 5.2% below 0.05, and 1.3% below 0.01—rates close to those expected under random assignment. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Emergency-cash receipt and conflict exposure by targeting arm

Targeting arm	Cash receipt	Conflict exposed	Mean events	Mean fatalities	N
PMT-Exp	0.567	0.561	14.2	6.2	1855
PMT-FCS	0.568	0.576	13.7	6.6	1842
PMT-CARI	0.541	0.571	13.4	6.4	1839
Expert	0.603	0.545	13.1	6.7	1863
Joint test (p)	0.003	0.234	0.610	0.766	

Notes: Share of households receiving WFP emergency cash and share in a conflict-exposed locality (any ACLED event, Oct 2024–Feb 2025), with mean events and fatalities, by targeting arm. The joint test is an F-test of equality across the four arms (robust SEs). Conflict exposure is balanced across the randomly assigned arms on every measure; emergency-cash receipt, realized after assignment, is modestly higher in the Expert arm (60% vs. 54–57%), a difference we address in the text.

Table A9: Pre-evaluation census (2020–2021) balance by targeting arm

	PMT-Exp	PMT-FCS	PMT-CARI	Expert	Joint p
FCS (full, 8-group)	33.14	33.35	33.51	33.47	0.861
rCSI	21.63	20.51	21.36	21.50	0.217
Sold domestic assets (LCSI stress)	0.351	0.377	0.378	0.363	0.386
Sold productive assets (LCSI crisis)	0.114	0.113	0.121	0.123	0.826
Household size	3.58	3.57	3.58	3.67	0.497
Dependency ratio	0.681	0.652	0.649	0.718	0.140
Age of household head	54.63	55.15	55.22	54.15	0.200
N	1,292	1,261	1,326	1,283	

Notes: Means by targeting arm at the pre-evaluation census (2020–2021), the earliest round in which the food-security module was collected for the study sample. *Joint p* is the F-test of equality across the four arms, with locality-clustered standard errors. Targeting arm is randomly assigned within eligibility strata; the balanced values confirm that the randomization held into the pre-conflict period used for the parallel-trends assessment.

Table A10: Conflict exposure and prior receipt of assistance

	Conflict intensity (z)		Any conflict	
	(1)	(2)	(3)	(4)
Prior in-kind receipt	-0.0180** (0.0083)	0.0437*** (0.0068)	0.0034 (0.0041)	0.0241*** (0.0034)
District fixed effects	No	Yes	No	Yes
R^2	0.000	0.446	0.000	0.407
Observations	58,501	58,501	58,501	58,501

Notes: Each column regresses a measure of locality conflict exposure on an indicator for prior receipt of WFP in-kind assistance, over the full beneficiary census ($N = 58,501$). The outcome is standardized ACLED conflict intensity ($\log(1 + \text{fatalities})$, Oct. 2024–Feb. 2025) in columns (1)–(2) and an indicator for any conflict event in columns (3)–(4); columns (2) and (4) add district fixed effects. The unconditional association is negligible—a 0.018 standard-deviation difference in intensity and an insignificant 0.3-percentage-point difference in any-conflict exposure—indicating that conflict exposure is essentially orthogonal to prior assistance. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A11: Characteristics of households selected by each targeting method

	PMT-Exp	PMT-FCS	PMT-CARI	Expert
Household size	4.72	4.03	4.53	3.20
Age of household head	51.74	54.75	52.97	62.02
Head married	75.6	65.9	70.8	41.2
Children under 15	1.34	1.04	1.25	0.47
No member employed	28.1	35.8	29.5	60.3
Elderly head (>64)	16.8	24.8	19.7	46.1
Working-age members (15–64)	3.03	2.56	2.89	1.99
Female-headed household	24.9	32.2	28.1	53.6
Elderly members (>64)	0.34	0.43	0.39	0.74
Monoparental household	16.0	21.2	18.1	39.6
Head: any disability	19.2	21.7	21.5	38.9
Female members	2.41	2.14	2.35	1.84
Children 0–5	0.42	0.33	0.39	0.15
Head: physical disability	12.2	14.9	14.3	26.7
Children 0–2	0.18	0.13	0.16	0.07
Head: completed primary	33.0	41.7	35.9	30.3
Dependency ratio	0.36	0.38	0.37	0.43
Assets index	-0.50	-0.65	-0.64	-0.41
Wealth index (z)	-0.47	-0.60	-0.59	-0.38
Head: incomplete primary	28.9	30.7	30.3	37.2
Households	25,156	25,156	25,156	25,160

Notes: Each column reports the mean of the characteristic among the 43% of households that the method selects for continued assistance, computed over the beneficiary census ($N = 58,501$). Binary indicators are in percent, continuous variables in their original units. Rows are the 20 characteristics with the largest cross-method spread—the range of the four selected-group means in population standard-deviation units—ordered from largest to smallest. Because a household can be selected by several methods, the selected groups overlap and across-method significance tests are not reported.

Table A12: Top predictors by contribution share, by targeting method

	PMT-Exp	PMT-FCS	PMT-CARI	Expert
1	Crowding, rooms/member (20.3)	Internet, WiFi/ADSL (21.2)	Vacuum cleaner (19.9)	Head unemployed (20.3)
2	Car (11.2)	Vacuum cleaner (14.7)	Crowding, rooms/member (16.2)	Married head (14.0)
3	Household-size quartile (11.1)	Car (13.9)	Car (9.8)	Any disabled member (11.8)
4	Household size (10.7)	Flush toilet (7.2)	Cooking oven (6.1)	Head disability (9.0)
5	Area per member (9.2)	Cooking oven (6.6)	Area per member (5.5)	Working-age males (7.8)
6	Children ≤ 15 (5.2)	Flat-screen TV (6.2)	Household-size quartile (4.8)	Female-headed HH (4.9)
7	Children, school age (4.1)	Microwave (5.5)	Household size (4.8)	Adults 15–64 (3.9)
8	Crowded dwelling, ≥ 1.5 (4.0)	Washing machine (5.4)	Flat-screen TV (4.6)	Working-age population (3.8)
9	Vacuum cleaner (3.3)	Air conditioner (3.1)	Microwave (4.1)	Age of head (3.7)
10	Household size, top quintile (2.4)	Head primary only (2.0)	Washing machine (4.0)	Senior head >64 (3.4)
Top-10 share	81.5%	85.8%	79.7%	82.7%

Notes: Predictors are ranked within each targeting model by their contribution—the absolute value of the penalized LASSO coefficient times the predictor’s standard deviation, normalized so that contributions sum to 100% within a model. Shares (%) are in parentheses; the final row reports the combined share of each model’s top ten predictors. The three PMT models load on assets, household size, and crowding, whereas the Expert model loads almost entirely on categorical vulnerability markers—unemployment, marital status, disability, and household composition.

Table A14: Progressivity: CGH index and targeting differential

Method	GT	CGH index			Targeting differential		
		10%	20%	43%	10%	20%	43%
PMT-Exp	Cons.	2.22	2.05	1.67	0.20	0.35	0.48
	FCS	1.53	1.38	1.29	0.09	0.13	0.21
	CARI	2.17	1.90	1.55	0.20	0.30	0.39
PMT-FCS	Cons.	2.03	1.83	1.60	0.17	0.27	0.43
	FCS	1.78	1.62	1.40	0.14	0.21	0.30
	CARI	1.98	1.79	1.52	0.16	0.26	0.38
PMT-CARI	Cons.	2.15	2.02	1.68	0.19	0.34	0.49
	FCS	1.35	1.42	1.34	0.06	0.14	0.25
	CARI	2.13	1.87	1.57	0.19	0.29	0.41
Expert	Cons.	1.23	1.32	1.32	0.04	0.11	0.23
	FCS	1.32	1.15	1.17	0.06	0.05	0.12
	CARI	1.20	1.19	1.24	0.03	0.06	0.17

Notes: CGH index (Coady et al., 2004) is the share of assistance reaching the deprived over their population share (equals 1 under random targeting; >1 is progressive). The targeting differential (Galasso and Ravallion, 2005) is the difference between the share of the deprived and of the non-deprived reached (ranges -1 to 1 ; 0 is non-progressive). Both are shown at the 10/20/43% targeted shares, against each ground truth (GT: consumption, FCS, CARI).

Table A13: Targeting accuracy by method and ground-truth outcome (full matrix)

Method	Spearman ρ	AUC	Precision	Recall
<i>Panel A. Ground truth: log per-capita consumption</i>				
PMT-Exp	0.57 [0.51, 0.62]	0.83 [0.80, 0.86]	0.69 [0.64, 0.75]	0.72 [0.67, 0.77]
PMT-FCS	0.46 [0.40, 0.52]	0.77 [0.73, 0.81]	0.64 [0.58, 0.69]	0.69 [0.63, 0.74]
PMT-CARI	0.52 [0.46, 0.58]	0.81 [0.77, 0.84]	0.67 [0.62, 0.73]	0.71 [0.66, 0.77]
Expert	0.29 [0.22, 0.36]	0.67 [0.63, 0.71]	0.55 [0.48, 0.60]	0.56 [0.50, 0.61]
Random	-0.00 [-0.08, 0.07]	0.50 [0.45, 0.54]	0.41 [0.35, 0.47]	0.43 [0.38, 0.47]
<i>Panel B. Ground truth: Food Consumption Score</i>				
PMT-Exp	0.25 [0.18, 0.32]	0.65 [0.61, 0.69]	0.54 [0.49, 0.60]	0.57 [0.50, 0.63]
PMT-FCS	0.35 [0.28, 0.41]	0.71 [0.66, 0.74]	0.58 [0.52, 0.63]	0.62 [0.56, 0.68]
PMT-CARI	0.29 [0.22, 0.36]	0.67 [0.63, 0.71]	0.57 [0.51, 0.62]	0.60 [0.54, 0.66]
Expert	0.17 [0.09, 0.24]	0.60 [0.55, 0.64]	0.49 [0.43, 0.54]	0.50 [0.44, 0.56]
Random	0.00 [-0.07, 0.08]	0.50 [0.46, 0.54]	0.42 [0.36, 0.48]	0.43 [0.38, 0.47]
<i>Panel C. Ground truth: CARI food insecurity</i>				
PMT-Exp	0.45 [0.38, 0.51]	0.77 [0.72, 0.80]	0.62 [0.57, 0.67]	0.68 [0.62, 0.74]
PMT-FCS	0.40 [0.32, 0.46]	0.74 [0.69, 0.77]	0.59 [0.53, 0.64]	0.66 [0.60, 0.71]
PMT-CARI	0.45 [0.39, 0.51]	0.77 [0.73, 0.80]	0.62 [0.56, 0.67]	0.69 [0.64, 0.75]
Expert	0.21 [0.13, 0.28]	0.62 [0.58, 0.67]	0.50 [0.44, 0.55]	0.54 [0.47, 0.59]
Random	-0.00 [-0.08, 0.07]	0.50 [0.46, 0.54]	0.40 [0.34, 0.45]	0.43 [0.38, 0.47]

Notes: Targeting accuracy on the held-out LHS fold. Spearman's ρ and the area under the ROC curve (AUC) are threshold-agnostic; precision and recall are evaluated at the 43% eligibility cut-off. Each method's continuous score is compared to the indicated ground-truth outcome. *Random* targets households at random and anchors the no-information benchmark. 95% bootstrap confidence intervals (1,000 replications) in brackets.

Table A15: Progressiveness: logit of selection on the asset index

	Asset index
PMT-Exp	-1.450*** (0.061)
PMT-FCS	-2.015*** (0.078)
PMT-CARI	-1.756*** (0.069)
Expert	-1.080*** (0.053)
Joint test of equal slopes (p)	0.000

Notes: Each cell is the coefficient on the welfare index from a separate logit of an indicator for selection by that arm on the index (robust SE in parentheses); negative values mean poorer households are more likely selected. The joint test reshapes to household-by-arm and tests equality of the index slope across the four arms (logit with arm interactions, SE clustered by household). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A16: Welfare gain by number of methods that select the household

Methods selecting the household	Households (N)	Share of caseload	Mean welfare gain (USD p.c.)	Per-capita exp. when cut (USD)
0 (never eligible)	380	0.265	8.89	112.13
1	1,536	0.190	10.51	119.91
2	2,308	0.157	9.89	105.70
3	1,531	0.238	9.56	94.12
4 (core / always)	392	0.149	10.20	91.06

Notes: Households grouped by how many of the four targeting methods deem them eligible (from the eligibility vector used for stratified randomization). *Mean welfare gain* is the survey×size-weighted per-capita benefit of continued assistance (negative discontinuation effect plus the in-kind parcel value); *per-capita expenditure when cut* is the baseline deprivation level. Core households (selected by all four methods) are the poorest—expenditure of \$91 against \$112–\$120 for the never- and least-eligible—so the methods agree on who is chronically deprived. The welfare gain from continued assistance, by contrast, is essentially flat across the groups (\$8.9–\$10.5 per capita, with no monotonic gradient): the never-eligible gain nearly as much as the core. The gain does not track deprivation or the number of selecting methods, so the marginal households over whom the methods disagree have gains close to the caseload average. Reallocating them across methods moves welfare little, which is why the four methods deliver near-identical welfare.

Table A17: Heterogeneous ITT effects on downstream outcomes by baseline subgroup

	FCS (1)	rCSI (2)	Food Exp. (USD) (3)	WHO-5 Index (4)
<i>Female-headed household (33% of households)</i>				
PMT-FCS	1.36 (1.01)	-1.23* (0.71)	7.96* (4.83)	1.66 (1.14)
PMT-CARI	2.05** (1.01)	-0.55 (0.72)	1.63 (4.94)	-0.19 (1.16)
Expert	0.72 (1.03)	-0.39 (0.72)	-0.68 (4.79)	0.30 (1.16)
<i>Elderly head (65+) (27% of households)</i>				
PMT-FCS	2.24* (1.19)	-1.27* (0.76)	18.61*** (5.82)	1.31 (1.27)
PMT-CARI	1.69 (1.18)	-0.10 (0.79)	3.51 (5.74)	0.65 (1.26)
Expert	2.57** (1.19)	-1.71** (0.75)	10.62* (5.57)	1.05 (1.24)
<i>Large household (5+ members) (34% of households)</i>				
PMT-FCS	-1.22 (1.01)	1.22 (0.79)	4.04 (4.80)	0.19 (1.14)
PMT-CARI	0.89 (1.01)	1.01 (0.81)	1.70 (5.08)	0.93 (1.17)
Expert	-1.34 (1.00)	1.23 (0.82)	2.58 (5.01)	-0.28 (1.14)
<i>Bottom asset quartile (25% of households)</i>				
PMT-FCS	1.36 (1.09)	-2.44*** (0.82)	15.62*** (5.32)	2.46* (1.28)
PMT-CARI	0.59 (1.09)	-0.18 (0.87)	1.34 (5.29)	0.35 (1.28)
Expert	0.85 (1.12)	-1.09 (0.85)	6.88 (5.13)	0.09 (1.28)
<i>Eligible (continued under assigned method) (50% of households)</i>				
PMT-FCS	-2.20*** (0.80)	-1.08* (0.63)	-4.83 (3.95)	0.39 (0.97)
PMT-CARI	0.05 (0.81)	0.03 (0.65)	0.17 (4.06)	-0.61 (0.97)
Expert	-1.61* (0.83)	-1.64*** (0.62)	-19.41*** (3.90)	-2.21** (0.93)
<i>Eligible under 3 methods (25% of households)</i>				
PMT-FCS	-0.40 (1.13)	-1.47* (0.89)	10.55* (5.44)	2.10 (1.35)
PMT-CARI	0.73 (1.16)	-0.06 (0.88)	3.78 (5.73)	-0.71 (1.34)
Expert	-0.64 (1.19)	-0.92 (0.88)	7.94 (5.53)	-0.06 (1.33)
<i>Eligible under 2 methods (37% of households)</i>				
PMT-FCS	0.58 (0.95)	-0.75 (0.73)	4.88 (4.84)	-0.35 (1.12)
PMT-CARI	1.21 (0.94)	0.38 (0.74)	-1.68 (4.79)	0.10 (1.13)
Expert	0.11 (0.95)	-0.32 (0.73)	3.35 (4.80)	-0.17 (1.10)
<i>Eligible under 1 method (25% of households)</i>				
PMT-FCS	-0.32 (1.17)	0.14 (0.85)	6.66 (5.97)	0.02 (1.34)
PMT-CARI	1.06 (1.21)	0.38 (0.89)	5.42 (5.99)	2.29* (1.33)
Expert	0.06 (1.17)	0.57 (0.85)	2.08 (6.03)	0.86 (1.31)

Notes: Each cell reports the arm×subgroup interaction coefficient from a regression of the downstream outcome on targeting arm, the baseline subgroup indicator, and their interaction (Round 1; PMT-Expenditure is the reference arm; control for the longer warning period). The coefficient is the differential effect of the arm for subgroup members relative to non-members. Subgroups are defined at baseline (pre-randomization). Eligibility (continued versus discontinued) is assigned by each method, so that panel blends targeting with the discontinuation margin. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A18: Impacts of Targeting on Downstream Outcomes by Eligibility (ITT)

	FCS	rCSI	Food Exp. (USD)	Total Exp. (USD)	LCSI Stress (pp)	LCSI Crisis (pp)	LCSI Emer- gency (pp)	WHO-5 Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Reference method: PMT-Exp</i>								
Panel A: Eligible (continued under assigned method)								
PMT-FCS	-2.20*** (0.80)	-1.08* (0.63)	-4.83 (3.95)	-30.33*** (10.39)	0.37 (1.18)	2.12 (2.48)	-0.00 (1.40)	0.39 (0.97)
PMT-CARI	0.05 (0.81)	0.03 (0.65)	0.17 (4.06)	4.58 (10.85)	0.82 (1.16)	3.46 (2.48)	0.97 (1.43)	-0.61 (0.97)
Expert	-1.61* (0.83)	-1.64*** (0.62)	-19.41*** (3.90)	-40.22*** (10.80)	1.17 (1.14)	5.02** (2.46)	-1.32 (1.34)	-2.21** (0.93)
Mean PMT-Exp.	55.24	17.45	135.36	392.93	93.88	55.25	8.50	23.73
Observations	3,200	3,200	3,200	3,200	3,200	3,200	3,200	3,166
Panel B: Not eligible (discontinued under assigned method)								
PMT-FCS	2.54*** (0.84)	-0.01 (0.60)	18.89*** (4.32)	64.04*** (11.65)	2.22** (1.08)	4.09* (2.48)	-0.50 (1.38)	0.32 (0.95)
PMT-CARI	2.05** (0.84)	0.63 (0.60)	4.29 (4.31)	8.31 (11.34)	0.26 (1.17)	-0.83 (2.49)	-0.08 (1.39)	0.86 (0.97)
Expert	1.53* (0.83)	1.73*** (0.62)	28.70*** (4.22)	67.69*** (11.25)	1.92* (1.09)	-6.14** (2.50)	-1.29 (1.34)	2.02** (0.96)
Mean PMT-Exp.	53.69	15.78	137.93	391.48	94.06	56.89	8.34	22.26
Observations	3,158	3,158	3,158	3,158	3,158	3,158	3,158	3,134

Notes: ITT effects of each targeting arm relative to PMT-Expenditure, estimated from a single regression per outcome that interacts the targeting arm with an indicator for whether the household was *eligible* (continued) under its assigned method (Round 1). Panel A reports the arm effect among eligible households; Panel B among not-eligible (discontinued) households. Because eligibility is itself assigned by each method, this split blends targeting with the discontinuation margin. Food and total expenditure are winsorized at the 95th percentile; LCSI coefficients and means are in percentage points; WHO-5 ranges 0–100. Robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A19: Impacts of Targeting on Downstream Outcomes with Weights (ITT)

	FCS	rCSI	Food Exp. (USD)	Total Exp. (USD)	Life Satisf. Index (SD)	Worriedness Index (SD)	WHO-5 Index
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
PMT-FCS	-0.33 (0.89)	0.06 (0.63)	6.35 (4.52)	9.79 (11.82)	-0.04 (0.05)	-0.05 (0.05)	-0.09 (1.02)
PMT-CARI	0.96 (0.88)	0.33 (0.62)	5.41 (4.41)	15.50 (12.01)	-0.05 (0.05)	-0.07 (0.05)	-1.27 (1.05)
Expert	-0.00 (0.87)	0.56 (0.64)	5.65 (4.49)	10.48 (11.86)	-0.05 (0.05)	-0.02 (0.05)	-1.42 (1.02)
Mean PMT-Exp.	54.47	16.62	136.64	392.21	-0.00	0.02	23.00
Observations	6,358	6,358	6,358	6,358	6,358	6,357	6,300
p-val(PMT-FCS=PMT-CARI)	0.151	0.660	0.831	0.633	0.861	0.787	0.233
p-val(PMT-FCS=Expert)	0.713	0.434	0.876	0.953	0.825	0.484	0.165
p-val(PMT-CARI=Expert)	0.275	0.716	0.956	0.676	0.964	0.336	0.882

Notes: PMT-Expenditure targeting serves as the reference group. The table shows the regression results on pre-specified primary outcomes at the household level using survey data. Each column is its own dependent variable. The independent variable of interest is the ITT treatment indicator and includes beneficiaries and non-beneficiary households. Regressions use sampling weights; robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A20: Average impact of conflict intensity

Outcome	Conflict intensity	N
<i>Physical harm</i>		
Displaced	0.225*** (0.022)	6455
Shelter damaged	0.178*** (0.021)	6458
<i>Psychological</i>		
Trauma (PTSD)	1.166*** (0.296)	6460
Mental well-being (WHO-5)	0.105 (0.718)	11060
<i>Asset depletion</i>		
Asset index	-0.144** (0.073)	12922
Sold productive assets	0.005 (0.021)	3053
<i>Coping</i>		
rCSI	-0.296 (0.552)	11180
LCSI stress (pp)	-0.881 (1.462)	11180
LCSI crisis (pp)	1.975 (1.729)	11180
LCSI emergency (pp)	1.243 (0.758)	11180

Notes: Effect of a one standard deviation increase in ACLED conflict intensity between Round 1 (mid-2024) and Round 2 (early 2025). Repeated outcomes (WHO-5, rCSI, LCSI severity, asset index) use a two-period difference-in-differences with household fixed effects (intensity in the post period); outcomes observed only after the escalation (displacement, shelter damage, trauma, productive-asset sales) use a Round 2 cross-section with eligibility-cell fixed effects. *N* counts household-by-period observations, so the difference-in-differences rows carry roughly twice the household count. LCSI severity indicators are in percentage points. Robust standard errors clustered at the locality level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A21: PMT-FCS heterogeneity in the conflict impact: full sample vs. continuers

Outcome (PMT-FCS $\times$ intensity)	Full sample	Continuers
<i>Physical harm</i>		
Displaced	-0.025* (0.015)	-0.075*** (0.023)
Shelter damaged	-0.008 (0.015)	-0.050** (0.024)
<i>Psychological</i>		
Trauma (PTSD)	0.255 (0.216)	-0.354 (0.387)
Mental well-being (WHO-5)	-0.102 (0.805)	1.470 (1.348)
<i>Asset depletion</i>		
Asset index	-0.026 (0.060)	-0.070 (0.121)
Sold productive assets	-0.046** (0.021)	-0.036 (0.037)
<i>Coping</i>		
rCSI	-0.473 (0.601)	-0.593 (0.856)
LCSI stress (pp)	-3.219* (1.827)	-1.393 (2.669)
LCSI crisis (pp)	-2.159 (2.322)	-2.023 (3.469)
LCSI emergency (pp)	-1.893 (1.249)	-1.035 (1.881)

Notes: PMT-FCS $\times$ conflict-intensity interaction (relative to PMT-Exp), by outcome, in the full sample and among households that continued receiving assistance. Conflict intensity is a one standard deviation increase in ACLED intensity between Round 1 (mid-2024) and Round 2 (early 2025). Outcomes observed only after the escalation (displacement, shelter damage, trauma, productive-asset sales) use a Round 2 cross-section with eligibility-cell fixed effects; repeated outcomes (WHO-5, asset index, rCSI, LCSI severity) use a two-period difference-in-differences with household fixed effects. LCSI severity indicators are in percentage points. Standard errors clustered at the locality level in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A22: Characteristics of selected households: impact-based vs. deprivation-based targeting

	PMT- Exp	PMT- FCS	PMT- CARI	Expert	Impact- Total	Impact- Food	Caseload
Household size	5.32	4.93	5.29	4.14	5.26	5.19	4.54
Head married	81.4	76.3	79.1	52.8	78.5	77.4	77.3
Children under 15	1.57	1.41	1.55	0.80	1.27	1.18	1.10
No member employed	25.0	29.0	24.9	53.2	25.4	26.0	24.3
Elderly head (>64)	14.4	19.4	16.3	41.1	32.7	26.2	19.8
Working-age members (15–64)	3.42	3.14	3.39	2.62	3.39	3.53	3.04
Female-headed household	21.4	24.9	22.8	45.0	23.6	23.9	23.6
Monoparental household	13.2	16.1	14.5	33.4	15.3	16.4	15.5
Head: any disability	16.0	17.2	17.2	36.7	17.8	19.5	15.1
Assets index	-0.45	-0.60	-0.55	-0.45	0.01	-0.11	0.03
Wealth index (z)	-0.39	-0.52	-0.47	-0.38	0.02	-0.08	0.03
Households (obs.)	3,049	3,109	3,078	3,077	2,270	2,282	6,147

Notes: Survey sample of 6,144 households (6,147 observations) with cross-fitted treatment effects. Cells are survey×household-size weighted means of baseline (pre-treatment) characteristics among the households each method selects; binary indicators in percent, continuous variables in their original units. The four deprivation/vulnerability arms (PMT-Exp, PMT-FCS, PMT-CARI, Expert) select households by their assigned eligibility category; the two impact arms rank households by predicted per-capita welfare loss from aid discontinuation (Impact-Total on the total-expenditure CATE, Impact-Food on the food-expenditure CATE) and keep the top 43% by weight. The Caseload column is the full sample.

Table A23: Welfare capture by targeting method (G_m)

Method	Total exp. ($\rho = 3$)	Food exp. ($\rho = 3$)
PMT-Exp	0.64 (0.02)	0.70 (0.02)
PMT-FCS	0.67 (0.02)	0.70 (0.02)
PMT-CARI	0.69 (0.02)	0.73 (0.02)
Expert	0.62 (0.02)	0.64 (0.02)
Impact-Total	0.91 (0.01)	0.83 (0.01)
Impact-Food	0.86 (0.01)	0.85 (0.01)
Random	0.52 (0.02)	0.54 (0.02)

Notes: $G_m = (V_m - V_0)/(V_{1B} - V_0)$ is the share of the attainable (first-best) welfare gain captured by method m , following [Aiken et al. \(2025\)](#) and [Hanna and Olken \(2018\)](#). Social welfare uses a CRRA aggregator over per-capita expenditure with $\rho = 3$; the first-best is evaluated at each arm’s own realized coverage. *Impact-Total* and *Impact-Food* rank households by their predicted per-capita welfare loss from discontinuation (cross-fitted treatment effects); *Random* anchors the no-information floor. Bootstrap standard errors (within eligibility strata) in parentheses.

Table A24: Why the deprivation methods deliver similar welfare: gain captured by each rule

Method	Corr. with welfare gain	Mean gain among selected (USD p.c.)	Lift over caseload	Share of first-best gain captured
PMT-Exp	-0.010	9.554	-0.156	-0.026
PMT-FCS	-0.042	9.654	-0.055	-0.009
PMT-CARI	0.029	9.697	-0.013	-0.002
Expert	0.156	11.093	1.384	0.234
Impact-Total	0.697	14.518	4.809	0.813
Impact-Food	0.436	12.750	3.041	0.514
Random	0.002	9.810	0.101	0.017
<i>Caseload mean</i>	—	9.710	—	—
<i>First-best</i>	—	15.625	5.915	1.000

Notes: The welfare gain is the per-capita benefit of *continued* assistance on total expenditure (the negative discontinuation effect plus the in-kind parcel value, as in the welfare engine). Column 1 is the rank correlation between a method’s selection and that gain; columns 2–4 give the survey×size-weighted mean gain among each rule’s selected 43%, its lift over the full-caseload mean, and the share of the first-best lift (a rule ranking directly on the gain) captured. The four deprivation methods barely sort on the gain (correlations near zero, and similar to one another) and cluster just above the caseload, which is why they deliver near-identical welfare; the impact arms sort on the gain and capture far more of it. Their moderate *welfare*-capture (G_m of 0.62–0.69 in Table A23) instead reflects their selection of poor households, whom the CRRA objective weights heavily.

Table A25: Welfare capture on well-being and conflict outcomes (G_m , $\rho = 3$)

Method	FCS	rCSI	Assets	Trauma	WHO-5
PMT-Exp	-1.93 (0.30)	0.49 (0.08)	0.31 (0.01)	0.31 (0.04)	0.32 (0.05)
PMT-FCS	-1.93 (0.26)	0.36 (0.07)	0.26 (0.01)	0.19 (0.03)	0.24 (0.05)
PMT-CARI	-2.11 (0.31)	0.37 (0.07)	0.28 (0.01)	0.20 (0.03)	0.26 (0.04)
Expert	-1.56 (0.24)	0.17 (0.04)	0.25 (0.01)	0.18 (0.03)	0.08 (0.04)
Impact-Total	-0.97 (0.14)	0.20 (0.04)	0.43 (0.02)	0.15 (0.03)	0.28 (0.05)
Impact-Food	-0.87 (0.15)	0.14 (0.03)	0.37 (0.02)	0.07 (0.02)	0.12 (0.03)
Random	-1.35 (0.22)	0.35 (0.08)	0.45 (0.02)	0.39 (0.04)	0.10 (0.05)

Notes: Welfare-capture ratio G_m (share of the first-best gain) on the five well-being and conflict outcomes, using a CRRA aggregator ($\rho = 3$) applied to the standardized outcome. The impact arms—ranked on predicted expenditure loss—do not lead here: on reduced coping (rCSI) the deprivation-based methods capture more, and trauma and mental well-being (WHO-5) are close to untargetable, with random allocation no worse than the rest. FCS is degenerate ($G_m < 0$): continued in-kind assistance does not raise the Food Consumption Score, so the first-best gain is not positive. Bootstrap standard errors in parentheses.

Table A26: Cost-effectiveness of targeting methods (total expenditure, $\rho = 3$)

Method	Raw G_m	C_m/cycle	% of B	ρ_m	G_m^{adj}
PMT-Exp	0.64	\$913,515	4.4	0.956	0.64
PMT-FCS	0.67	\$913,515	4.4	0.956	0.67
PMT-CARI	0.69	\$913,515	4.4	0.956	0.68
Impact-Total	0.91	\$913,515	4.4	0.956	0.91
Impact-Food	0.86	\$913,515	4.4	0.956	0.87
Expert (registry K=2)	0.62	\$615,160	3.0	0.970	0.62
Expert (registry K=3)	0.62	\$465,982	2.2	0.978	0.62
Expert (registry K=5)	0.62	\$316,805	1.5	0.985	0.62
Expert (full re-survey)	0.62	\$913,515	4.4	0.956	0.62
Random	0.52	\$0	0.0	1.000	0.51

Notes: Welfare capture on per-capita total expenditure (CRRRA $\rho = 3$), before and after folding each method's per-cycle data-collection cost into the transfer budget (Aiken et al., 2025). The annual transfer budget is $B \approx \$20.7\text{M}$, derived from the parcel rule (\$32/month, one parcel per two members capped at three) over the 43% included. **Assumed unit costs:** a household survey costs \$15 per household and the analyst \$12,000 per month for three months (\$36,000 per cycle); between refreshes the registry pays maintenance equal to 15% of a full survey ($\phi=0.15$); costs are spread over a $T=5$ -year horizon and discounted at 3%. The PMTs, the impact arms, and Expert-full re-survey all $N=58,501$ households each cycle (\$877,515 in survey cost plus the \$36,000 analyst, or \$913,515), while the Expert-registry re-surveys only every K cycles and pays maintenance in between (averaged over the horizon). $\rho_m = (B - C_m)/B$ is the budget-retention factor; G_m^{adj} recomputes welfare capture with the parcel scaled by ρ_m . Random screening cost is zero.